EC325 Workbook

EC325: Econometrics | Colby College

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Preface

About these Notes

This booklet contains the lecture notes for EC 325: Econometrics at Colby College. These notes are meant to supplement the course readings.

The content in this book is *heavily inspired* by three main texts, which are required for this course:

1. Wooldridge (2019)
2. Angrist and Pischke (2014)
3. Wickham, Çetinkaya-Rundel, and Grolemund (2023)
4. Bailey (2020)

Structure of These Notes

This course is basically broken into three modules:

1. Correlation, Causation, and Regression
2. Real-Life Regression Models
3. Introduction to Causal Methods

Each chapter contains both notes for the course - in narrative form - as well as econometric labs that we will work through in R.

Roadmap

- In Unit 1, we will discuss the difference between correlation and causality, and what it means for us as economists.
- In Unit 2 we will learn about the "gold standard" of causal inference–Randomized Control Trials (RCTS). We will also be introduced to the Potential Outcomes Framework, a mathematical formalization of distinguishing correlation from causality.

Statistical Software

Almost all modern econometrics is done with specialized statistical software which you interact with via code.

About Me

I am a recent PhD from UMass Amherst. My research focuses on the relationship between public/economic policy and maternal, infant, and child health and racial/ethnic health disparities. You can find out more about me and my work on my website: www.raymondcaraher.com

1 Introduction

Key Questions

- What is the difference between correlation and causation?
- What does an econometrics research project entail?
- What are the main types of data formats we will work with?
- What is statistical software and why is it useful?

Suggested Readings

- Wooldridge (2019), Ch. 1
- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1

1.1 What is this Class About?

This class is obviously dedicated to the study of econometrics (endearingly shortened 'metrics). But what exactly does econometrics entail? What differentiates it from broader terms such as "statistical analysis" or "data analysis"?

From Wooldridge (2019), we get a classic definition of econometrics:

> "...statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy."

However, my view of econometrics is at the same time, broader and more narrow:

Definition of Econometrics

The practice of using statistical, quantitative methods to study social phenomena and provide insights into public policy. More specifically, modern econometrics has focused on identifying the **causal effects** of social phenomena/policy as opposed to **correlations**.

In other words, I view the practice of econometrics as the use of quantitative methods to understand the social world around us, with a special (though not exclusive) lens focusing on separating correlation from causation.

1.2 Does Having Health Insurance *Cause* You to be Healthier?

Let's think about a critical question in public policy: *does having health insurance make someone healthier*? This is especially an important question in the context of the United States, which does not have a public health insurance system. Instead, it is a predominantly private system entangled with a complex net of social safety nets meant to provide a patchwork of coverage to the most marginalized.

However, as of 2023, 9.5% of U.S. adults do not have any health insurance coverage.[1] Would the US as a whole be better off guarantee health insurance coverage for everyone? If health insurance makes people healthier, then there may be a strong case to do so. If not, the benefits of such a system would become, at the very least, more opaque.

Let's take a look at life expectancy in the US (a course but important measure of population health), relative to similar high-income, developed countries. Figure 1.1 shows the average life expectancy from 1990 to 2021 for the US, UK, Canada, Australia, and a number of other high-income countries. As can be seen, the US performs considerably worse in terms of life expectancy, and this divergence between peer countries has gotten *larger* since the 1990s, and especially during the COVID-19 pandemic.

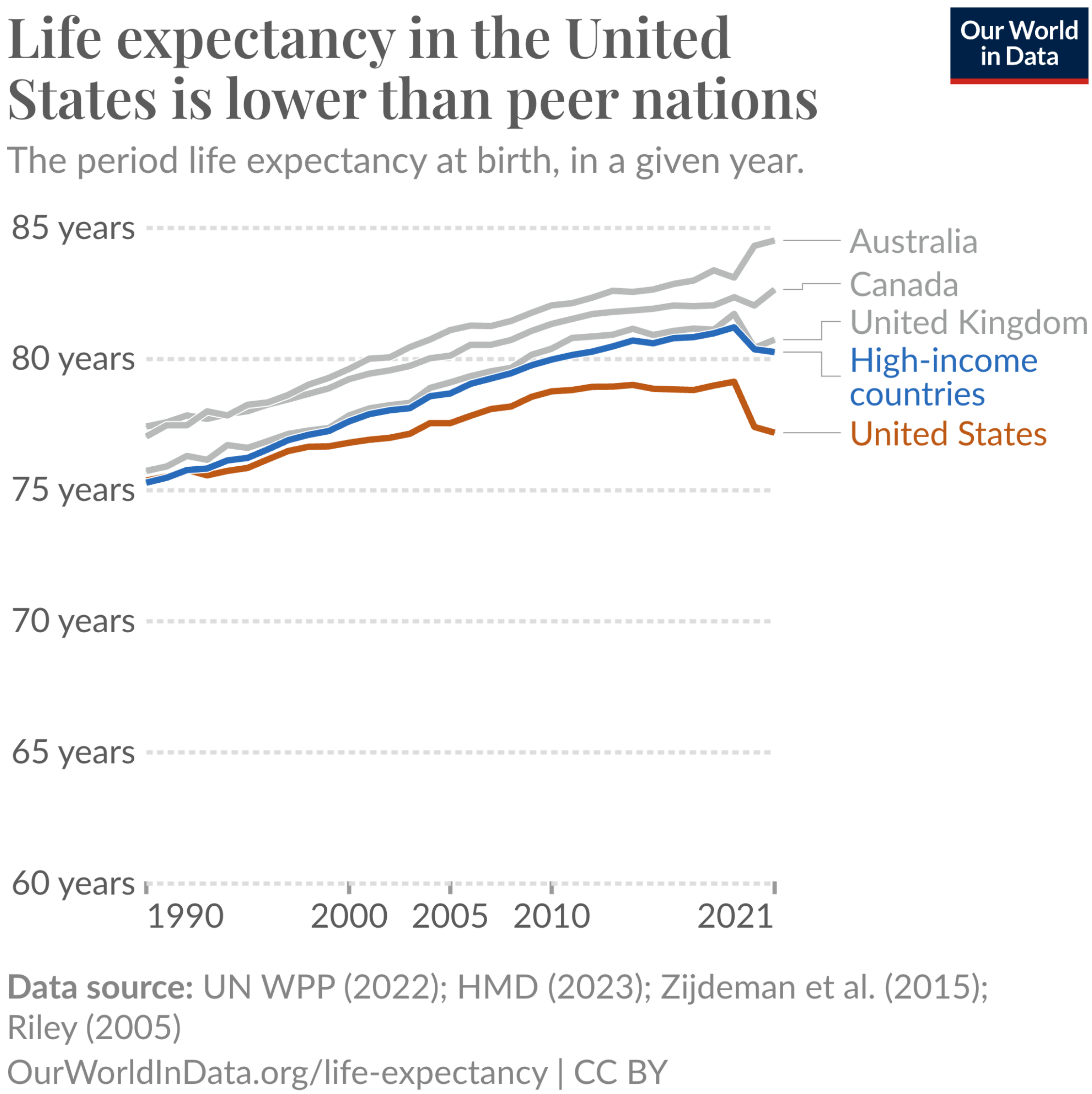
Critically—with the exception of the US—all of the countries in the peer group have some form of universal health care. In other words, all developed countries which are healthier than the US have policies which guarantee the uninsured rate is zero.

This then leads to a critical question: did universal health insurance *cause* these countries to have better health outcomes? Or is this a correlation between policy and outcomes? Certainly, one could make an argument that having health insurance does cause better health outcomes. On the other hand, there are lots of differences between the US and other high-income countries which may also explain the pattern seen in Figure 1.1. For example, maybe these countries have different regulations regarding the food quality, which may explain part of the difference in life expectancy. Or maybe because relatively more people commute to work via bicycle or walking, which may have health benefits compared to driving to work.

These other differences between groups which may *confound* our comparison between countries with universal health insurance and those without also extends to individuals. In the US at least, those with health insurance may be more likely to have a high-paying job which includes insurance as a benefit. Therefore, a comparison between those who have insurance and those who do not will be plagued with the fact that those who have insurance *also are more likely*

[1] Source: KFF

Figure 1.1: Life expectancy in the US vs. peer countries.



Source: Our World in Data

to have a high income job, who may have had better health outcomes regardless of if they had insurance or not. In other words, we may be confusing *correlation* for **causation*.

> **Correlation vs. Causation**
>
> The **correlation** between two variables describes how they move together. It means two things are *associated*. **Causation** describes a direct relationship where a change in one variable causes a change in another. It means one thing *makes the other happen.*

A key part of the content in this course is to think about when our comparisons are describing a **correlative** relationship or a **causal** relationship. We will study techniques which—if one is convinced that the requisite assumptions hold—will actually be able to recover the *causal effect* of social phenomena, and bring important insights into essential public policy questions like: what is the effect of health insurance on health outcomes? Does going to college cause you to have higher wages? Do immigrants reduce native wages? and does policing cause decreases in crime?

1.3 Broad Steps of Econometric Analysis

Let's say we want to know what the effect of on-the-job training has on your wage. What steps would we need to take to answer this question?

1.3.1 Step 1: Write the Model

Our first step would be to write an *econometric model,* with the goal of using data and econometric tools to *estimate the parameters of the model.* This model can sometimes be derived from a formal model in economics (i.e., the profit maximization of firms), but it is at least based on some intuitive logic which mathematically relates some variable (the **dependent variable**) to some **independent variable**(s).

An example of the type of econometric model you may want to run to study the effect of on-the-job training is:

$$wage = \beta_0 + \beta_1(training) + \mu, \tag{1.1}$$

where *wage* is the *outcome, left-hand side (LHS), predicted, or dependent variable* (these are synonyms for the same thing), and *training* is the *explanatory, right-hand side (RHS), predictor, or independent variable* (again, synonyms for the same thing). The β_0 and β_1 variables are the parameters that we want to *estimate the value of* using econometric methods, and they tell us *how* our *independent* variable is related mathematically to our dependent variable.

1.3.2 Step 2: Find and Explore the Data

The next thing you have to do—after we figure out the model we want to estimate, such as in Equation 1.1—is to get the data. There is a *huge* range of datasets out there, and we are especially lucky that the data-gathering infrastructure in the United States is incredibly robust. As we go through the semester, we will work with as many different datasets that we can, which cover a range of topics: from sports to health, crime, and economic development. While different datasets usually can their own quirks, broadly speaking there are three different types of data we will work with in this class:

1. **Cross-sectional data:** This is data which contains information on units (e.g., individuals, countries, firms, households, etc.) at a *single point in time.* This data is the most common, and also is generally the simplest to work with, so we will start out by working mostly with cross-sectional data.

2. **Time-series data:** This data consists of observations on a single unit (e.g., a country's Gross Domestic Product (GDP), a specific company's stock price, or the national unemployment rate) across multiple time periods (e.g., years, quarters, months, or even minutes). In time series data, a crucial characteristic is the temporal ordering of the observations. This structure is often used for forecasting and modeling how a variable's past values influence its present and future values.

3. **Panel (or Longitudinal) data**: Panel data combines both cross-sectional and time-series dimensions. It consists of observations on multiple units (e.g., individuals, firms, or countries) over multiple time periods. It essentially tracks the same set of cross-sectional units over time. This gives it more information, more variability, and allows researchers to use a rich set of methods to really hone in on the causal effects of whatever social phenomena being studied.

In all of these cases, econometric data is organized in what is called a **matrix** or a **dataframe**. It should look roughly similar to an Excel Spreadsheet, where each column represents a different variable (which is usually labelled, sometimes not), and each row represents the values that a specific unit has across these variables.

For example, the data we use to estimate the model in Equation 1.1 may look like Table 1.1,

where the **person_name** column is (obviously) each person's name, **person_id** column corresponds the unique numerical identifier associated with each person (this will be more common to see in datasets since names are usually anonymous), the **wage** column shows each person's hourly wage in dollars, and the **yrs_training** column shows the number of years each person received on-the-job training. This row-column structure of organizing data is so fundamental that the word *column* is often interchanged with the word *variable*.

When you first look at some data, it is really important to figure out what each row represents. In the simple example shown in Table 1.1, this is straightforward since each row represents a different person. However, in more complex datasets, it may not be immediately obvious what

Table 1.1: Relationship between training and wages for 10 workers

person_name	person_id	wage	yrs_training
Paul	1	18.50	1.00
Ringo	2	22.00	3.00
George	3	17.65	2.50
John	4	27.95	8.20
Linda	5	19.20	0.50
Yoko	6	24.80	6.80
Pattie	7	20.50	5.50
Maureen	8	23.75	4.20
Cynthia	9	16.40	0.80
Barbara	10	28.50	9.10

each row represents. For example, in panel data, each row may represent a specific person at a specific point in time. So row 1 may be Paul in year 2020, row 2 may be Paul in year 2021, row 3 may be Ringo in year 2020, and so on. Figuring out what each row represents is *critical* to understanding the data, as well as how to appropriately analyze it.

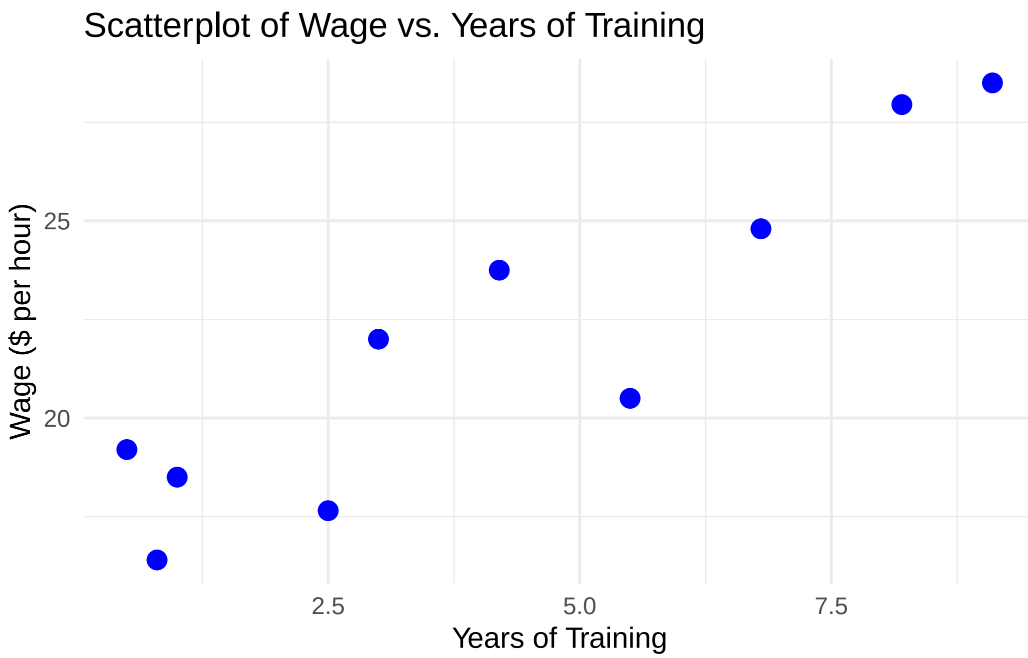
Once we have the data in hand, then we may need to do some data cleaning and organization. This may involve: - Removing missing values - Creating new variables - Changing the format of variables (e.g., from string to numeric) - Merging datasets together - Filtering the data to only include certain observations - And many other tasks.

These are all tasks that in many cases take up the bulk of the time in an econometric research project, and while not the most glamorous part of econometrics, it is absolutely essential to get right.

Before moving on to estimation, it is also important to do some **exploratory data analysis**, (which should be considered step 2.5 in the econometric research process). This involves looking at summary statistics of the data (means, medians, standard deviations, etc.), as well as visualizations (histograms, scatter plots, etc.). Exploratory data analysis is important for understanding the data, as well as for identifying potential issues with the data (e.g., outliers, missing values, etc.).

One of the most important parts of exploratory data analysis is to look at the relationship between the dependent variable and independent variable(s) in your model. One way this can be done is using a **scatterplot**, which plots each observation in the dataset using the independent and dependent variables as coordinates, with the independent variable on the x-axis and the dependent variable on the y-axis.

Figure 1.2: Relationship between training and wages for 10 workers



In Figure 1.2, we see a scatterplot of the relationship between years of training and wages for the 10 workers shown in Table 1.1. From this plot, we can see that there is a positive relationship between years of training and wages: as years of training increases, wages also tend to increase. While not good enough to draw any strong conclusions, one can see how this type of exploratory data analysis is useful for understanding the data and the relationship between variables.

1.3.3 Step 3: Estimate the Model

Once we have a model in mind as well as the appropriate data, we can then use our econometric methods to estimate the relationship. The bulk of this class is dedicated to this step in the econometric research process, but the other stages are no less important! The most common way that econometric models are estimated is with **Ordinary Least Squares** (OLS), which you may remember from your pre-req. In addition to estimating the actually value of the parameters in the model, we also want some idea of how *uncertain* we are about the estimated parameter value. This is where notions of statistical significance and confidence intervals come in.

1.3.4 Step 4: Interpret the Estimate

Once we have an estimated number for the parameter in our model, we then have to interpret the number. Here, the phrase "interpret the estimate" means two parallel things. First, there is the literal, mathematical interpretation of the model. This will change depending on the functional form we use to estimate the model, and we will spend lots of time learning about this. The second part of "interpreting" the model is about what the results mean in the context of your broader research question. Is your estimate big or small? Is it economically important? Do you actually think it reflects a causal effect? These are all types of interpretation which are a critical part of econometrics research, but for which there is not usually a formula we can use. A large part of the "art" working on an econometrics project is exactly this type of interpretation, and how you form it into a compelling narrative. While there are some tools we will cover which will help in this regard, in my opinion, nothing will help you more with this than just reading lots of good, well-written econometrics papers.

1.4 Statistical Software

Much of the econometric process is done using specialized statistical software. We will use this software to: "clean" our data and organize in the appropriate row-column format to do our econometric estimations; explore our data using descriptive statistics (means, medians, etc.) and visualizations (an increasingly important part of a compelling econometric project); estimate our model and conduct analyses of statistical uncertainty; and output our results to easy-to-interpret tables and figures.

There are a few commonly used econometrics software:

- **Stata:** Proprietary software purpose-built for econometrics
- **R:** Open-source programming language with a focus on statistical analysis
- **Python:** Open-source programming language with a focus on data science more broadly
- **EViews:** Proprietary software with a focus on forecasting and other time series applications

Each one of these software of their own pros and cons, and it is worth knowing several quite well. For the purposes of this class, we will learn the **R** programming language to do econometrics. If you haven't take the time to review Appendix A to learn more about R, how to install it, and to run your first chunk of code!

1.5 Summary and Conclusion

This chapter introduced the fundamental framework that will guide our study of econometrics throughout this course. We defined econometrics as the use of quantitative methods to understand social phenomena and inform public policy, with a particular focus on distinguishing between correlation and causation. Through the example of health insurance and health outcomes, we saw how patterns in data can be misleading. While the United States lags behind peer countries in life expectancy, and those peer countries all have universal health insurance, we cannot immediately conclude that universal coverage causes better health outcomes. The tools you will learn in this course are designed to help you identify causal relationships rather than mere associations.

1.5.1 Key Takeaways:

- **Econometrics vs. correlation:** Modern econometrics focuses on identifying causal effects, not just correlations between variables
- **Four steps of econometric research:**
    1. Write an econometric model
    2. Find and explore the appropriate data
    3. Estimate the model parameters
    4. Interpret the results
- **Three main data types:** Cross-sectional (single point in time), time-series (one unit over time), and panel data (multiple units over time)
- **Statistical software:** We will use **R** to clean data, estimate models, and present results

As we move forward, remember that econometrics is as much an art as it is a science. The mathematical tools we will learn are powerful, but they are only as good as the questions we ask and the care with which we interpret our results.

1.6 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

2 Causality and Randomized Control Trials

Key Questions

- Why can't we interpret correlations as causal effects?
- What is selection bias and how does it arise?
- How do randomized control trials eliminate selection bias?
- What is the potential outcomes framework and how does it formalize causal inference?

Suggested Readings

- Angrist and Pischke (2014), Ch. 1

In Chapter 1, we discussed how econometrics focuses on identifying **causal effects** rather than mere correlations. We used the example of health insurance: while people with health insurance tend to be healthier than people without insurance, we cannot immediately conclude that health insurance *causes* better health. This chapter develops the conceptual and mathematical tools we need to think rigorously about causality, and introduces the randomized control trial (RCT) as the gold standard for causal inference.

2.1 The Problem with Simple Comparisons

Let's return to our health insurance example. People with health insurance are, on average, healthier than people without it. This correlation is undeniable. But can we interpret this as evidence that insurance *causes* better health?

The problem is that people who have health insurance differ from those who don't in many ways beyond just their insurance status. Those with insurance tend to be wealthier, more educated, and have more stable employment. All of these factors independently affect health outcomes. When we compare the health of insured and uninsured individuals, we're not just comparing the effect of insurance—we're comparing people who differ along many dimensions.

Selection Bias

Selection bias occurs when we compare groups that differ systematically in ways beyond just the treatment we're studying. In other words, we are comparing **apples to oranges**.

The comparison we *want* to make is between two otherwise identical people who differ only in whether they have health insurance. In other words, we want an **apples to apples** comparison. This is the essence of the *ceteris paribus* condition: **all else being equal**, the difference between treated and untreated units reveals the causal effect of treatment.

2.2 The Multiverse of Possibilities

To build intuition for causal inference, consider the television show *Rick and Morty*. In this show, the character Rick possesses a "portal gun" that allows him to travel across infinite parallel universes. In each universe, there exists a version of his grandson Morty who is identical in every way except for one characteristic—Cowboy Morty, Evil Morty, Hammer Morty, and so on. Each Morty has the same home, same parents, same sister, same *everything else*.

This multiverse provides the perfect setting for causal inference! Comparisons between different Mortys are truly *ceteris paribus* comparisons because everything except the one characteristic of interest is held constant.

2.2.1 The Causal Effect of College

Suppose we want to know the causal effect of attending college on wages. We can't simply compare the wages of college graduates to non-graduates because these groups likely differ in many unobserved ways (motivation, family background, innate ability, etc.).

But imagine we could observe two versions of Morty from parallel universes: Morty C-137 (who didn't attend college) and College Morty (who did). These two Mortys are identical in every way—same intelligence, same family, same opportunities—except that one went to college and the other didn't.

The difference in wages between College Morty and Morty C-137 would be a genuine *ceteris paribus* difference—the true causal effect of college on wages.

Figure 2.1: Rick and Morty: A show about infinite parallel universes.

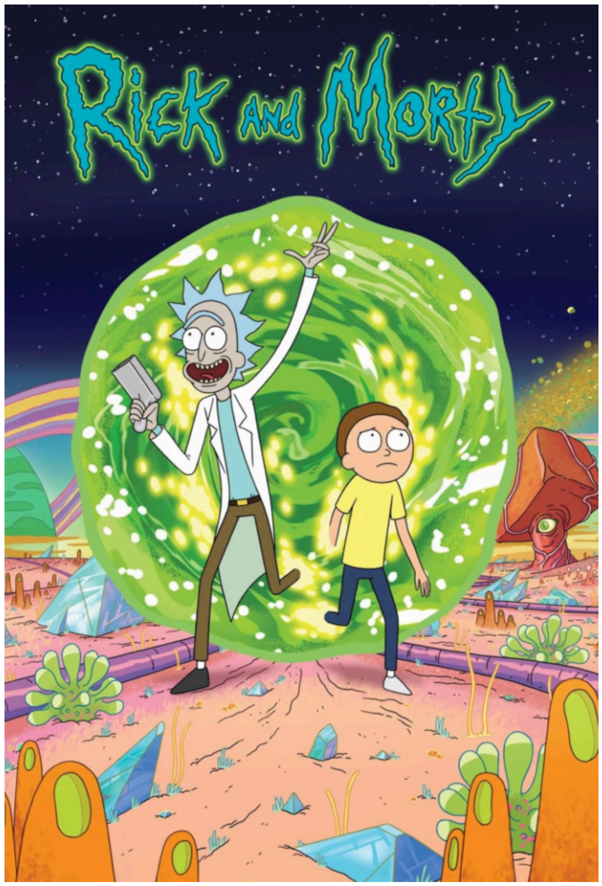


Figure 2.2: Morty C-137 (left) and College Morty (right). The wage difference between them represents the true causal effect of college.

(a) Morty C-137

(b) College Morty

2.2.2 The Fundamental Problem

Of course, the fundamental problem of causal inference is that **we don't have a portal gun**. We only observe one version of reality, not parallel universes. We can never observe what would have happened to a college graduate had they *not* gone to college, or what would have happened to a non-graduate had they attended. This unobserved "what if" scenario is called the **counterfactual**.

2.3 Randomized Control Trials: The Next Best Thing

Since we can't travel between parallel universes, we need another way to make valid causal comparisons. The solution that comes closest to achieving *ceteris paribus* conditions in the real world is the **randomized control trial** (RCT).

Consider how a pharmaceutical company tests whether a new drug works. They don't just give the drug to sick people and see if they get better—that would be subject to all sorts of selection bias. Instead, they conduct an RCT:

1. **Sample** a group of volunteers from the population of interest
2. **Randomly assign** some volunteers to receive the drug (the "treatment" group) and others to receive a placebo (the "control" group)
3. **Compare** the average outcomes between the treatment and control groups

The key insight is that **random assignment eliminates selection bias**. When treatment is assigned randomly, there is no systematic relationship between who receives treatment and their underlying characteristics. On average, the treatment group and control group will be identical in terms of age, health status, income, motivation, and every other characteristic—both observed and unobserved.

> **!** Why Randomization Works
>
> Random assignment ensures that, on average, the treatment and control groups have identical characteristics. Any difference in outcomes between the groups can therefore be attributed to the treatment itself, not to pre-existing differences between the groups.

2.4 The Law of Large Numbers

Why exactly does randomization eliminate selection bias? The answer lies in a fundamental result from probability theory: the **Law of Large Numbers** (LLN).

The LLN states that as the sample size increases, the sample average converges to the population average. Formally, if $Y_1, Y_2, ..., Y_n$ are independent, identically distributed random variables with mean μ, then:

$$\text{plim}(\bar{Y}_n) = \mu$$

where $\bar{Y}_n$ is the sample average and plim denotes the probability limit (what the value converges to as $n \to \infty$).

When we randomly assign individuals to treatment and control groups, we are essentially drawing random samples from the same underlying population. The LLN guarantees that, with a large enough sample, the average characteristics of each group will converge to the population average—and therefore to each other.

2.4.1 A Simulation

> **What is a Simulation?**
>
> A **simulation** is a computer-based experiment where we generate artificial data according to rules we specify. Unlike real-world data, we know exactly how the simulated data was created—including the true causal effects, which variables influence which outcomes, and how treatment was assigned.
> Simulations are powerful learning tools because they let us see what *should* happen under ideal conditions. If we set the true causal effect to zero and then compare treated and untreated groups, any difference we observe must be due to something other than a real treatment effect (like selection bias or random chance). By manipulating features of the simulation—such as sample size or how treatment is assigned—we can build intuition for how these factors affect our ability to recover causal effects. Think of simulations as a laboratory where we can run controlled experiments on statistical methods themselves.

To see the Law of Large Numbers in action, consider a simulation where we have a population with three observable characteristics: light-colored eyes, dark hair, and being tall. Suppose these characteristics are correlated with both treatment status and outcomes in observational data (i.e., without random assignment). Critically, we set the true causal effect of treatment on the outcome to **zero**.

> **A Note on the Code Below**
>
> The R code in the following sections is more advanced than what we've covered so far in the course. Don't worry if you don't understand every line! The code is provided for those who are curious and want to see how simulations work "under the hood." Even

if the syntax is unfamiliar, reading through the comments can help you understand the logic of what we're doing. As you progress through the course, more of this code will start to make sense.

Show Code: Generating the Simulated Data

```
# Load the tidyverse package for data manipulation and plotting
library(tidyverse)

# Set a "seed" for reproducibility
# This ensures we get the same random numbers each time we run the code
set.seed(181247)

# Define the TRUE causal effect of treatment
# We're setting this to ZERO so we know any observed difference is bias!
causal_effect <- 0

# Create a population of 10,000 individuals
n_population <- 10000

# Generate three random characteristics for each person
# Each characteristic is randomly 0 or 1 (like a coin flip)
eye_light <- sample(x = c(1, 0), size = n_population, replace = TRUE)
hair_dark <- sample(x = c(1, 0), size = n_population, replace = TRUE)
big_feet <- sample(x = c(1, 0), size = n_population, replace = TRUE)

# Generate random "noise" - unexplained variation in outcomes
err <- rnorm(n_population)

# Create an underlying "propensity" for treatment
# Notice: people with light eyes are MUCH more likely to be treated (coefficient = 2.1)
# People with dark hair are moderately more likely (coefficient = 0.83)
# Big feet has a small negative effect (coefficient = -0.11)
treatment_propensity <- 1 + 2.1 * eye_light + 0.83 * hair_dark - 0.11 * big_feet + err

# Assign treatment based on this propensity (treated if propensity > 2.25)
# This creates SELECTION BIAS: treatment is correlated with characteristics
treated <- ifelse(treatment_propensity > 2.25, 1, 0)

# Create the outcome variable
# The outcome depends on the same characteristics that predict treatment!
# But notice: we add "treated * causal_effect" which equals zero
# So treatment has NO real effect on the outcome
outcome <- 1 + 2.2 * eye_light + 1.2 * hair_dark - 0.01 * big_feet + treated * causal_effec

# Combine everything into a data frame
dat <- tibble(
  outcome = outcome,
  treated = treated,
  eye_light = eye_light,
  hair_dark = hair_dark,
  big_feet = big_feet
)

# Let's see how many people ended up in each group
cat("Number in each group:\n")
```

Number in each group:

```r
count(dat, treated)
```

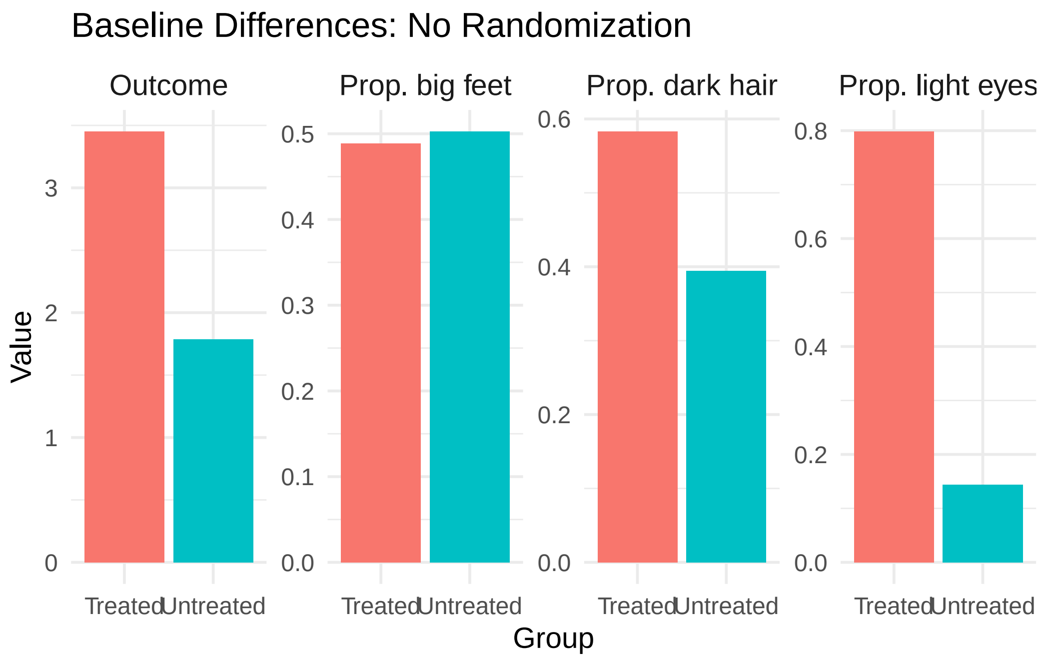
```
# A tibble: 2 x 2
  treated     n
    <dbl> <int>
1       0  4624
2       1  5376
```

When we simply compare treated and untreated individuals *without* random assignment, we see large differences between groups—not just in outcomes, but in their baseline characteristics. The treated group has more people with light eyes, more people with dark hair, and so on. These differences in characteristics lead to a misleading difference in outcomes, even though the true causal effect is zero.

Show Code: Visualizing Baseline Differences (No Randomization)

```
# Calculate the mean of each variable for treated vs untreated groups
baseline_diffs <- dat |>
  # Group the data by treatment status
  group_by(treated) |>
  # Calculate summary statistics for each group
  summarise(
    Outcome = mean(outcome),
    `Prop. light eyes` = mean(eye_light),
    `Prop. dark hair` = mean(hair_dark),
    `Prop. big feet` = mean(big_feet)
  ) |>
  # Reshape data from "wide" to "long" format for plotting
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
    values_to = "value"
  ) |>
  # Create nice labels for the treatment groups

  mutate(group = ifelse(treated == 1, "Treated", "Untreated"))

# Create the bar plot
ggplot(baseline_diffs) +
  geom_bar(
    aes(x = group, y = value, fill = group),
    stat = "identity"
  ) +
  # Create separate panels for each variable
  facet_wrap(~variable, nrow = 1, scales = "free_y") +
  labs(
    x = "Group",
    y = "Value",
    title = "Baseline Differences: No Randomization"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(size = 11)
  )
```

Figure 2.3: Baseline differences between treated and untreated groups without randomization. Despite the true causal effect being zero, we observe large differences in both characteristics and outcomes.



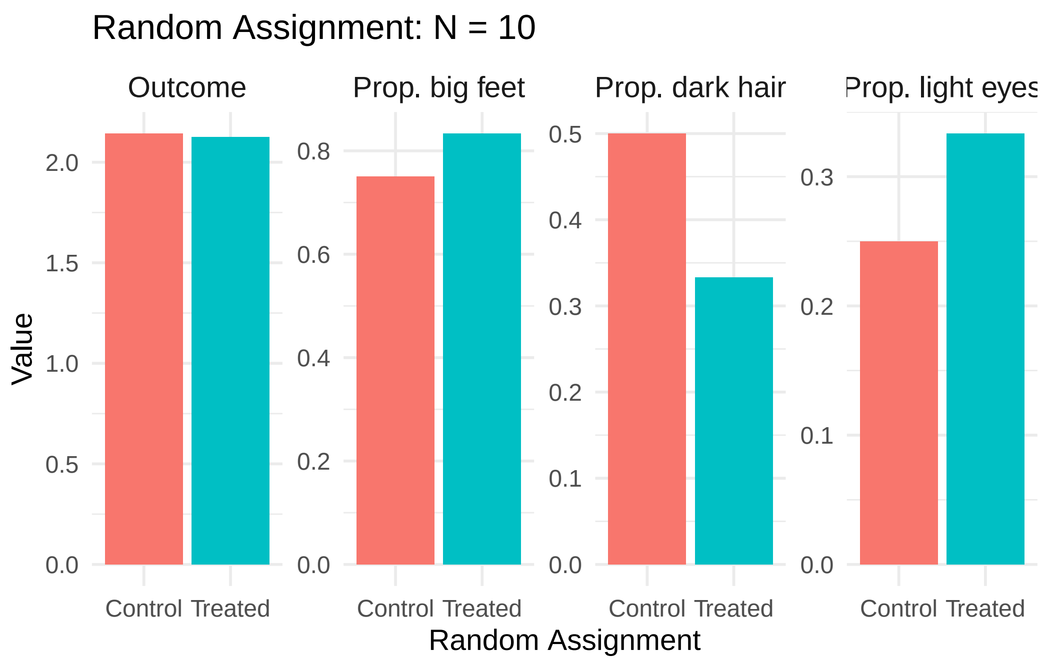
Notice how the treated and untreated groups look very different! The treated group has a higher proportion of people with light eyes and dark hair. Most importantly, the treated group has a higher mean outcome—even though **we know the true causal effect is zero**. This is selection bias in action: people who "select into" treatment have characteristics that also lead to better outcomes.

Now let's see what happens when we randomly assign treatment. Starting with a small sample (N=10), random assignment reduces but doesn't eliminate the differences between groups. With such a small sample, random chance can still produce imbalanced groups.

Show Code: Random Assignment Simulation

```r
# Function to run the random assignment simulation for a given sample size
run_random_assignment <- function(data, sample_size, seed = 181247) {

  # Set seed for reproducibility
  set.seed(seed)

  # Take a random sample from the untreated population
  # (We use untreated only to avoid any prior treatment effects)
  sample_data <- data |>
    filter(treated == 0) |>
    slice_sample(n = sample_size)

  # RANDOMLY assign treatment - this is the key step!
  # Each person has a 50% chance of being assigned to treatment
  # This assignment is INDEPENDENT of their characteristics
  random_treatment <- sample(c(1, 0), size = sample_size, replace = TRUE)

  # Add the random treatment assignment to our sample
  sample_data <- sample_data |>
    mutate(rand_treat = random_treatment)

  # Calculate means for each randomly assigned group
  sample_data |>
    group_by(rand_treat) |>
    summarise(
      Outcome = mean(outcome),
      `Prop. light eyes` = mean(eye_light),
      `Prop. dark hair` = mean(hair_dark),
      `Prop. big feet` = mean(big_feet),
      .groups = "drop"
    ) |>
    mutate(sample_size = sample_size)
}

# Run the simulation for different sample sizes
results_n10 <- run_random_assignment(dat, sample_size = 10)
results_n25 <- run_random_assignment(dat, sample_size = 25)
results_n50 <- run_random_assignment(dat, sample_size = 50)
results_n100 <- run_random_assignment(dat, sample_size = 100)

# Combine all results
all_results <- bind_rows(results_n10, results_n25, results_n50, results_n100)

# Reshape for plotting
all_results_long <- all_results |>
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
    values_to = "value"
  ) |>
  mutate(
```

Figure 2.4: Random assignment with N=10. Some imbalance remains due to small sample size.



With only 10 people, we still see noticeable differences between the randomly assigned treatment and control groups. This is not selection bias—it's just random chance with a small sample.

As we increase the sample size to N=25 and N=50, the differences between treatment and control groups shrink. The LLN is at work: larger samples produce more balanced groups.

By N=100, the treatment and control groups are nearly identical across all characteristics. The difference in outcomes between groups is now close to zero—the true causal effect! Any remaining difference is just random variation, not systematic selection bias.

2.5 The Potential Outcomes Framework

Now that we intuitively understand why simple comparisons can be misleading and how randomization addresses this problem, let's formalize these concepts mathematically using the **potential outcomes framework**.

Figure 2.5: Random assignment with N=25 (top) and N=50 (bottom). As sample size increases, the groups become more balanced.

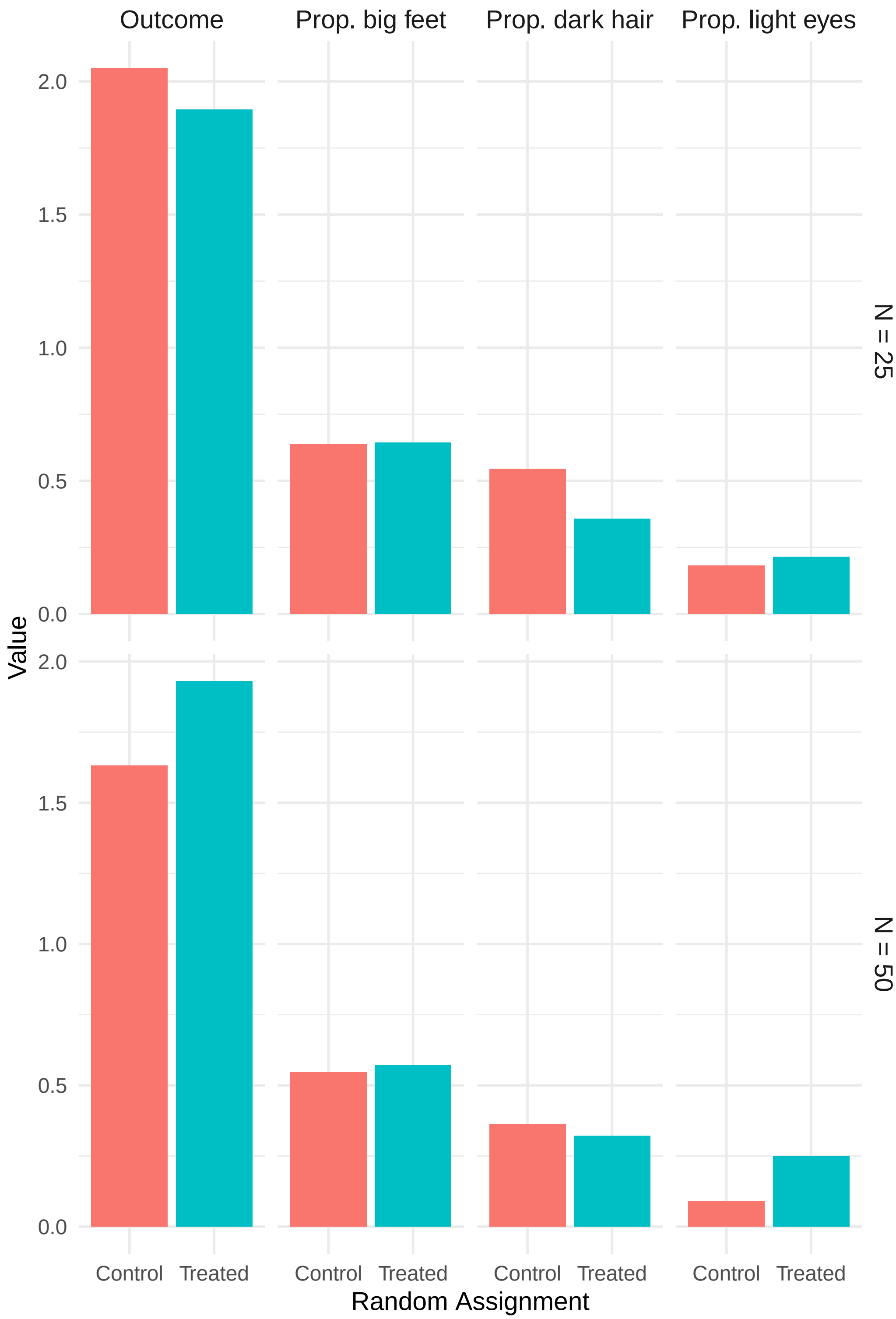
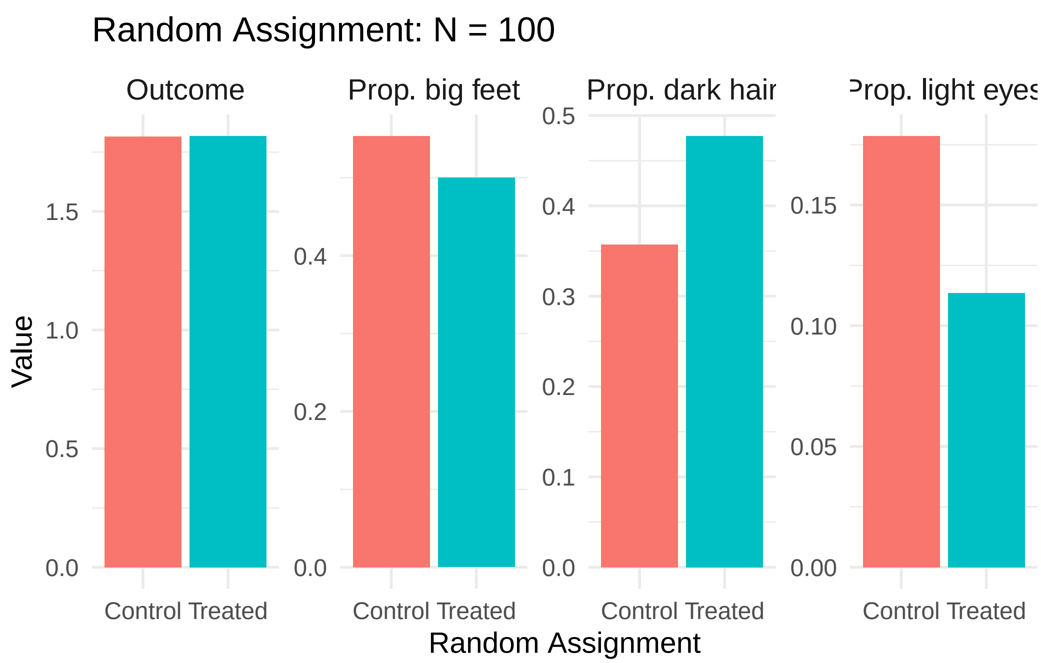


Figure 2.6: Random assignment with N=100. The treatment and control groups are now nearly identical in their characteristics, and the difference in outcomes is close to the true causal effect of zero.



2.5.1 Defining Potential Outcomes

Consider our example of estimating the causal effect of college on wages. For any individual i, we can define two **potential outcomes**:

- Y_{1i}: The wage individual i would earn **if they attend college**
- Y_{0i}: The wage individual i would earn **if they do not attend college**

The causal effect of college for individual i is simply the difference between these potential outcomes:

$$\kappa_i = Y_{1i} - Y_{0i}$$

This framework captures the "multiverse" intuition: Y_{1i} is College Morty's wage, Y_{0i} is Morty C-137's wage, and κ_i is the causal effect of college for Morty.

Figure 2.7: The potential outcomes framework conceptualizes causal effects as the comparison between two paths at a fork in the road.

2.5.2 The Fundamental Problem of Causal Inference

The fundamental problem is that we only observe **one** potential outcome for each individual. If person i attends college, we observe Y_{1i} but never see Y_{0i}. If they don't attend, we observe Y_{0i} but never see Y_{1i}. We never observe both potential outcomes for the same individual.

Let D_i be a treatment indicator:

$$D_i = \begin{cases} 1 & \text{if individual } i \text{ is treated (attends college)} \\ 0 & \text{otherwise} \end{cases}$$

What we actually observe is:

$$Y_i = D_i \cdot Y_{1i} + (1 - D_i) \cdot Y_{0i}$$

That is, we observe the treated outcome for treated individuals and the untreated outcome for untreated individuals—but never both.

2.5.3 Selection Bias Formalized

Now we can formally see why simple comparisons are problematic. Suppose we compare average outcomes between those who are treated ($D_i = 1$) and those who are untreated ($D_i = 0$). The difference in group means is:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$$

Since we only observe Y_{1i} for the treated and Y_{0i} for the untreated, this becomes:

$$E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0]$$

Here's the key insight. Let's add and subtract a term that represents the counterfactual—what the treated group's outcome would have been without treatment:

$$E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0] + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 1]\}}_{=0}$$

Rearranging terms:

$$\underbrace{E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1]}_{\text{Causal effect for the treated}} + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\}}_{\text{Selection bias}}$$

The Selection Bias Decomposition

A simple difference in group means equals the causal effect **plus** selection bias:

$$\underbrace{E[Y_i|D_i = 1] - E[Y_i|D_i = 0]}_{\text{Observed difference}} = \underbrace{\kappa}_{\text{Causal effect}} + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\}}_{\text{Selection bias}}$$

The selection bias term compares the **untreated potential outcomes** of the treated group to the untreated potential outcomes of the control group. In the health insurance example, this captures differences in baseline health (absent insurance) between those who choose to get insurance and those who don't. If people who get insurance would be healthier even without insurance (perhaps because they're wealthier or more health-conscious), then selection bias is positive, and the simple comparison overstates the causal effect.

2.5.4 How Randomization Eliminates Selection Bias

When treatment is randomly assigned, the treatment and control groups are drawn from the same population. By the Law of Large Numbers, their average characteristics—including their average potential outcomes—converge to each other. Formally:

$$E[Y_{0i}|D_i = 1] = E[Y_{0i}|D_i = 0]$$

This says that the average untreated potential outcome is the same for both groups. Plugging this into our decomposition:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = \kappa + \{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\} = \kappa + 0 = \kappa$$

With random assignment, the simple difference in means equals the causal effect. Selection bias disappears.

2.6 The Oregon Health Insurance Experiment

Most econometric research cannot rely on randomization to identify causal effects. We typically work with observational data where treatment is not randomly assigned, and much of this course focuses on methods for addressing selection bias in such settings. However, there have been a few landmark studies that used randomization to study important policy questions. One of the most influential is the **Oregon Health Insurance Experiment**.

2.6.1 Context: Medicaid Expansion

Throughout the 2000s, many U.S. states expanded their Medicaid programs to cover more low-income residents. Most states simply extended coverage to everyone who became newly eligible. But in 2008, Oregon did something unique: facing limited resources, they offered Medicaid coverage through a public lottery.

2.6.2 The Lottery

Oregon's approach created a natural randomized experiment. People who wanted coverage could sign up for the lottery. Winners received the opportunity to apply for the Oregon Health Plan (the state's Medicaid program), while losers remained uninsured. To be eligible, applicants had to be legal Oregon residents aged 19-64, uninsured for the past six months, with income below the federal poverty line, and not otherwise eligible for health insurance.

Because lottery winners were chosen randomly, the comparison between winners and losers provides a valid causal estimate of the effect of health insurance—free from the selection bias that plagues observational studies.

2.6.3 Results

Researchers followed lottery participants and found that lottery winners (relative to losers) experienced:

1. **Increased take-up of Medicaid:** The lottery actually changed insurance status, confirming the treatment "worked"
2. **Increased use of medical resources:** Winners used more hospital and outpatient services
3. **Small improvements in overall health:** Physical health improvements were modest, though mental health and financial security improved substantially

These findings are important because they come from a rigorous RCT rather than observational comparisons. The study provides some of the cleanest evidence we have on what health insurance actually *causes*, as opposed to what it merely correlates with.

Discussion Question

The Oregon experiment found modest effects of insurance on physical health. Does this mean expanding health insurance is not worthwhile? What other outcomes might matter for policy decisions?

2.7 Summary and Conclusion

This chapter introduced the fundamental concepts underlying causal inference in econometrics. We began with the intuition that simple comparisons between treated and untreated groups can be misleading due to selection bias—the groups may differ in ways beyond just the treatment. The "multiverse" thought experiment illustrated what we really want: *ceteris paribus* comparisons where everything except the treatment is held constant.

Randomized control trials approximate this ideal by randomly assigning treatment, which ensures that treatment and control groups are comparable on average. The Law of Large Numbers guarantees that with large enough samples, random assignment eliminates selection bias.

We then formalized these ideas using the potential outcomes framework, which defines causal effects as the difference between what would happen with and without treatment. The fundamental problem is that we only observe one potential outcome per individual. Simple comparisons of means capture both the causal effect and selection bias, but random assignment makes selection bias vanish.

2.7.1 Key Takeaways

- **Selection bias** arises when treated and untreated groups differ systematically beyond just the treatment—we're comparing apples to oranges
- **Ceteris paribus** ("all else equal") comparisons isolate the causal effect by holding everything constant except the treatment
- **Randomized control trials** eliminate selection bias by ensuring treatment and control groups are drawn from the same population
- The **Law of Large Numbers** guarantees that with large samples, randomly assigned groups have identical average characteristics
- The **potential outcomes framework** formalizes causal effects as $\kappa = Y_{1i} - Y_{0i}$, but we only observe one potential outcome per individual
- Simple differences in means equal: Causal effect + Selection bias. Randomization sets selection bias to zero.

2.8 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

3 Bivariate Regression and Ordinary Least Squares

Key Questions

- How do we mathematically model the relationship between two variables?
- What is the population regression model and how do we interpret its parameters?
- How do we derive the OLS estimator?
- What assumptions are required for OLS to be unbiased?
- What is omitted variable bias and when does it arise?
- How do we measure the precision of our estimates?

Suggested Readings

- Wooldridge (2019), Ch. 2

In Chapter 2, we learned that randomized control trials provide the gold standard for causal inference by eliminating selection bias through random assignment. But most economic questions cannot be answered with experiments. We typically work with observational data where "treatment" is not randomly assigned. This chapter introduces **Ordinary Least Squares (OLS)** regression, the workhorse tool of econometrics. We focus here on the **bivariate** case—a single independent variable—to build intuition before extending to multiple variables.

3.1 Modeling Relationships Between Variables

Suppose we have two variables, y and x. In econometrics, we are often interested in one of several related goals: estimating the **causal effect** of x on y, explaining variation in y using x, or simply describing how y varies with x. OLS is a tool that can help with all of these—though as we'll see, whether we can interpret results causally depends on assumptions that may or may not hold.

But what do we even *mean* when we say we want to study "how x is related to y"? We need a precise, mathematical way to operationalize this question. The most straightforward approach is to ask: what is the *average* value of y for each different value that x can take?

Consider some examples. How can we quantify the gender pay gap? We compare the *average* wage for men versus women. How do we measure the returns to education? We compare the *average* wage of people with different levels of education. In both cases, we are asking about how the average of one variable changes across values of another.

3.1.1 Conditional Expectations

This idea of finding "the average value of y for a given value of x" is exactly what we call the **conditional expectation**. Formally, the conditional expectation of y given x is written as:

$$E[Y|X = x]$$

This notation asks: if we knew that X took on some specific value x, what would we expect Y to be, on average? The conditional expectation is often the central quantity we want to model and estimate in econometrics.

3.1.2 Example: Education and Wages

Suppose X represents education level, which can take values $\{hsdeg, colldeg, profdeg\}$ (high school degree, college degree, professional degree), and Y represents hourly wage. For each education level, we can compute the average wage among people in that group.

The red points in Figure 3.1 show the conditional expectations—the average wage within each education category. These averages tell us how wages vary with education, which is precisely what we set out to measure.

3.2 The Population Regression Model

Now that we understand what we want to estimate (the conditional expectation), we need a framework for *how* to estimate it. Before we can estimate anything, we must first describe the underlying relationship we believe exists in the population.

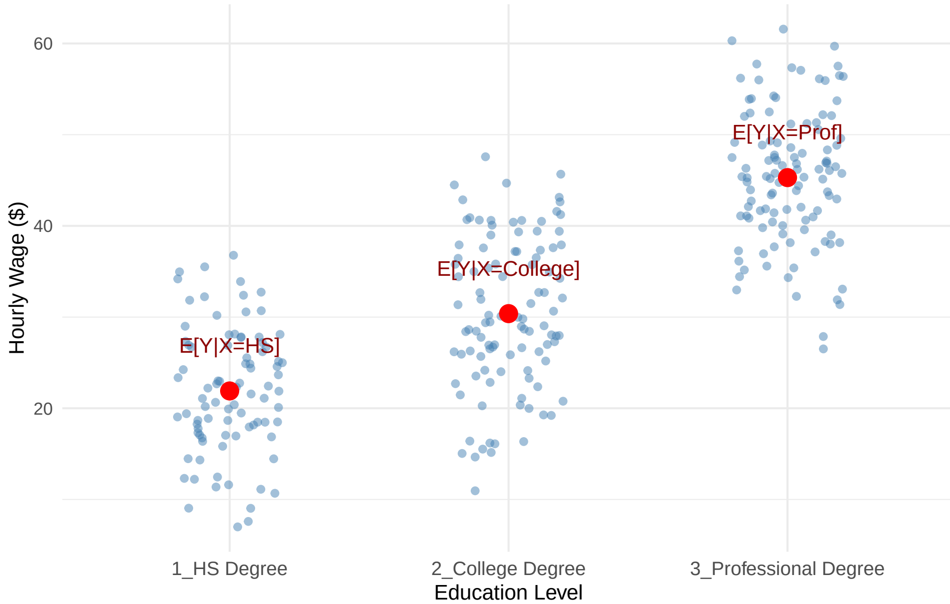
3.2.1 The Bivariate Model

The simplest specification is called the **bivariate** (or simple) regression model:

$$y = \beta_0 + \beta_1 x + \mu \tag{3.1}$$

where:

Figure 3.1: Wages by education level. Each point represents an individual. Red points show the conditional expectation (average wage) for each education group.



- y is the **dependent variable** (also called the explained, response, or outcome variable)
- x is the **independent variable** (also called the explanatory, control, or predictor variable)
- μ is the **error term**, which represents all unobserved factors that affect y
- β_0 and β_1 are **population parameters**—the "true" values describing the relationship between x and y in the population

> **The Error Term**
>
> The error term μ is crucial. It captures everything that affects y other than x. In a wage equation, μ might include ability, motivation, family connections, luck, and countless other factors we cannot observe or measure.

3.2.2 Connecting to Conditional Expectations

How does this regression model relate to our goal of estimating $E[Y|X]$? If we assume that the error term has mean zero conditional on x—that is, $E[\mu|x] = 0$—then:

$$E[y|x] = E[\beta_0 + \beta_1 x + \mu|x] = \beta_0 + \beta_1 x + E[\mu|x] = \beta_0 + \beta_1 x$$

This is exactly the conditional expectation we set out to estimate! The regression line $\beta_0 + \beta_1 x$ traces out the average value of y for each value of x.

3.2.3 Interpreting the Coefficients

The coefficients in Equation 3.1 have specific interpretations:

- β_0 **(the intercept):** The predicted value of y when $x = 0$. Depending on the context, this may or may not have a meaningful interpretation.
- β_1 **(the slope):** The **marginal effect** of x on y. Mathematically, it is the derivative:

$$\frac{dy}{dx} = \beta_1$$

This tells us: when x increases by one unit, y changes by β_1 units, on average.

3.2.4 Example: Education and Wages

A regression model relating years of education to hourly wages might be:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

where $wage$ is dollars per hour, $educ$ is years of education, and μ captures all other determinants of wages (experience, ability, etc.).

If $\beta_1 = 2.50$, this would mean that each additional year of education is associated with $2.50 higher hourly wages, on average.

3.3 The Regression Line: Predicted Values and Residuals

Before we derive estimators for β_0 and β_1, let's introduce some key concepts that will guide our approach.

3.3.1 Fitted Values and the Regression Line

Once we have estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, we can construct the **OLS regression line** (or fitted line):

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

For any value of x, this line gives us $\hat{y}$, the **predicted value** (or **fitted value**) of y. The regression line represents our best guess for the average value of y at each level of x.

3.3.2 Residuals: The Prediction Errors

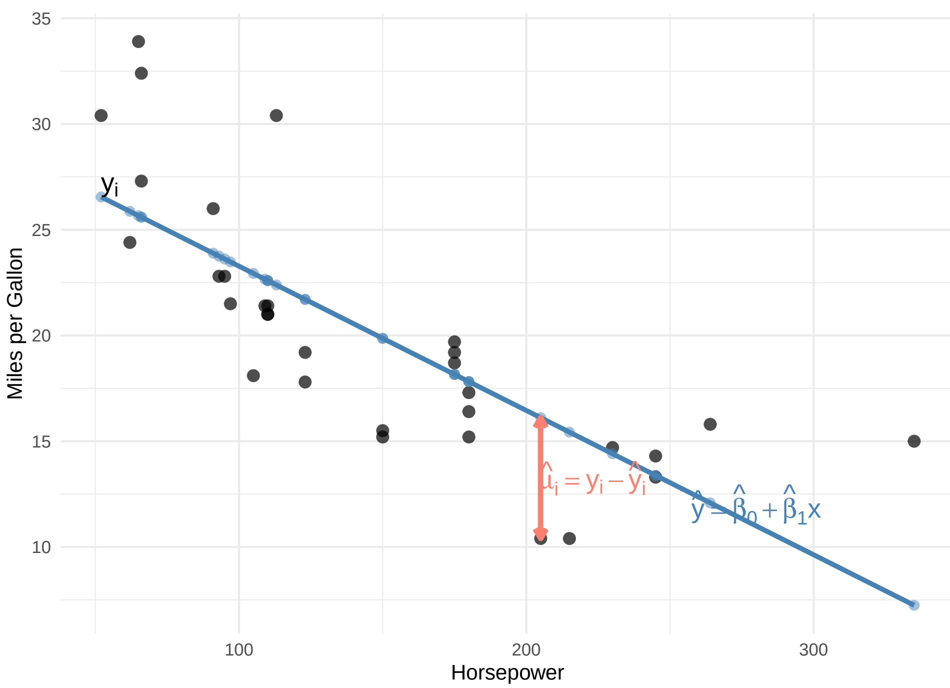
Of course, our predictions won't be perfect. The **residual** for observation i is the difference between the actual value and the predicted value:

$$\hat{\mu}_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

The residual measures how much our prediction "misses" for each observation. Positive residuals mean we underpredicted; negative residuals mean we overpredicted.

```
Warning in geom_segment(aes(x = x_point, y = y_actual, xend = x_point, yend = y_predicted),
i Please consider using `annotate()` or provide this layer with data containing
  a single row.

`geom_smooth()` using formula = 'y ~ x'

Warning in is.na(x): is.na() applied to non-(list or vector) of type
'expression'

Warning in is.na(x): is.na() applied to non-(list or vector) of type
'expression'

Warning in is.na(x): is.na() applied to non-(list or vector) of type
'expression'
```

Figure 3.2: Components of the regression line. Black points show actual values (y), the blue line shows predicted values ($\hat{y}$), and the salmon segment illustrates a residual ($\hat{u}$ = y - $\hat{y}$).



3.3.3 A Numerical Example

Let's see these concepts with actual numbers. The table below shows the first six observations from a regression of MPG on horsepower, along with the predicted values and residuals:

```
Warning: 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")

Warning: 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

Table 3.1: Example regression data showing actual values, predicted values, and residuals.

	x (Horsepower)	y (MPG)	ŷ (Predicted)	û (Residual)
Mazda RX4	110	21.0	22.59	-1.59
Mazda RX4 Wag	110	21.0	22.59	-1.59
Datsun 710	93	22.8	23.75	-0.95
Hornet 4 Drive	110	21.4	22.59	-1.19
Hornet Sportabout	175	18.7	18.16	0.54
Valiant	105	18.1	22.93	-4.83

Notice how the residual column equals the actual value minus the predicted value. For the first observation, the car has 110 horsepower and gets 21 MPG. Our regression predicts it should get about 22.59 MPG, so the residual is $21 - 22.59 = -1.59$. The negative residual means we *overpredicted*—the car actually gets slightly worse fuel economy than our model predicted.

3.3.4 The Key Insight: Minimize the Residuals

Here's the crucial idea: we want to choose $\hat{\beta}_0$ and $\hat{\beta}_1$ so that our predictions are as good as possible. In other words, we want to make the residuals as small as possible.

But we can't simply minimize the sum of residuals $\sum \hat{\mu}_i$, because positive and negative residuals would cancel out. Instead, we minimize the **Sum of Squared Residuals (SSR)**:

$$SSR = \sum_{i=1}^{n} \hat{\mu}_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

This is exactly where the name "Ordinary Least Squares" comes from: we find the estimates that make the sum of squared residuals as small as possible.

3.4 Deriving the OLS Estimator

Now we turn to the mechanics of estimation. The **population** regression model describes the true relationship, but we never observe the entire population or the true values of β_0 and β_1. Instead, we have a **sample** of data and must *estimate* the population parameters.

3.4.1 The Minimization Problem

Suppose we have a random sample of n observations, $\{(x_i, y_i) : i = 1, 2, ..., n\}$. As discussed above, we want to find the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squared residuals:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

This is a calculus optimization problem. The SSR is a function of two unknowns ($\hat{\beta}_0$ and $\hat{\beta}_1$), and we want to find the values that minimize it.

3.4.2 First-Order Conditions

To find the minimum, we take partial derivatives with respect to each parameter, set them equal to zero, and solve. These are called the **first-order conditions**.

Partial derivative with respect to $\hat{\beta}_0$:

$$\frac{\partial SSR}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Dividing by -2 and rearranging:

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Partial derivative with respect to $\hat{\beta}_1$:

$$\frac{\partial SSR}{\partial \hat{\beta}_1} = -2 \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Dividing by -2:

$$\sum_{i=1}^{n} x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

3.4.3 Solving for the Intercept

From the first condition, distributing the summation:

$$\sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{n} x_i = 0$$

Dividing by n and recognizing sample means:

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Solving for the intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{3.2}$$

This tells us that the regression line always passes through the point $(\bar{x}, \bar{y})$—the "center" of the data.

3.4.4 Solving for the Slope

Substituting Equation 3.2 into the second first-order condition and working through the algebra (using summation properties from Appendix B):

$$\sum_{i=1}^{n} x_i[y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i] = 0$$

After distributing and rearranging:

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n} x_i(x_i - \bar{x})$$

Using the summation properties $\sum x_i(y_i - \bar{y}) = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $\sum x_i(x_i - \bar{x}) = \sum(x_i - \bar{x})^2$:

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n}(x_i - \bar{x})^2$$

Solving for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{3.3}$$

3.4.5 Confirming It's a Minimum

How do we know this critical point is a minimum rather than a maximum? We check the second-order conditions. The second partial derivative of SSR with respect to $\hat{\beta}_1$ is:

$$\frac{\partial^2 SSR}{\partial \hat{\beta}_1^2} = 2\sum_{i=1}^{n} x_i^2 > 0$$

Since this is positive (as long as there's variation in x), the SSR function is convex, and our solution is indeed a minimum.

> **!** The OLS Slope Formula
>
> The OLS slope estimate equals the sample covariance of x and y divided by the sample variance of x:
>
> $$\hat{\beta}_1 = \frac{Cov(x, y)}{Var(x)} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$
>
> This can also be written in terms of the correlation coefficient:
>
> $$\hat{\beta}_1 = \rho_{xy}\frac{\sigma_y}{\sigma_x}$$

This last expression reveals something important: bivariate OLS is essentially a scaled version of the correlation between x and y. And as we discussed in Chapter 1, correlation does not imply causation! We must be very careful about interpreting bivariate OLS results causally.

3.5 Computing OLS in R

While we derived the formulas by hand, in practice we use software to compute OLS estimates. In R, the `lm()` function (for "linear model") performs OLS regression.

3.5.1 Basic Syntax

```r
# General syntax
lm(y ~ x, data = your_data)
```

The formula `y ~ x` specifies that y is the dependent variable and x is the independent variable. An intercept is included by default.

3.5.2 Example: MPG and Horsepower

Let's estimate the relationship between fuel efficiency (MPG) and engine horsepower using the built-in `mtcars` dataset:

```r
# Load the data
data(mtcars)

# Estimate the regression
lm_mpg <- lm(mpg ~ hp, data = mtcars)

# View the results
summary(lm_mpg)
```

```

Call:
lm(formula = mpg ~ hp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.7121 -2.1122 -0.8854  1.5819  8.2360 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
hp          -0.06823    0.01012  -6.742 1.79e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared:  0.6024,	Adjusted R-squared:  0.5892 
F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

The output shows our estimates: $\hat{\beta}_0 = 30.10$ and $\hat{\beta}_1 = -0.07$. The slope suggests that each additional unit of horsepower is associated with about 0.07 fewer miles per gallon.

3.6 Goodness-of-Fit: R-Squared

How well does our regression line fit the data? The **R-squared** (R^2) provides one measure of goodness-of-fit.

Define:

- **Total Sum of Squares (SST):** $SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$ — total variation in y
- **Explained Sum of Squares (SSE):** $SSE = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$ — variation explained by the regression
- **Residual Sum of Squares (SSR):** $SSR = \sum_{i=1}^{n}\hat{\mu}_i^2$ — unexplained variation

The R-squared is:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

This measures the fraction of the total variation in y that is explained by the regression. R^2 ranges from 0 (no explanatory power) to 1 (perfect fit).

> ⚠ Interpreting R-Squared
>
> Be careful interpreting R^2, especially in the context of causal inference. A high R^2 does not mean we have a causal estimate—it just means x is correlated with y. Conversely, a low R^2 does not mean our estimate is wrong or unimportant. In cross-sectional data, R^2 values of 0.1 to 0.3 are common and can still represent economically meaningful relationships.

3.7 The Gauss-Markov Assumptions

We've derived the OLS formulas, but how do we know if the estimates are any good? The formulas will always give us *some* numbers, but whether those numbers have useful statistical properties depends on certain conditions being met.

The **Gauss-Markov assumptions** are a set of conditions under which OLS has desirable properties. Understanding these assumptions is crucial: they tell us when we can trust our estimates and when we should be skeptical.

3.7.1 Assumption 1: Linear in Parameters

> **Gauss-Markov Assumption 1: Linear in Parameters**
>
> The population model is: $y = \beta_0 + \beta_1 x + \mu$

This assumption states that the dependent variable y can be written as a linear function of the parameters β_0 and β_1, plus an error term.

What this does NOT mean: This assumption is less restrictive than it might first appear. It does *not* require that y and x have a linear relationship. We can model many non-linear relationships by transforming the variables. For example:

- $y = \beta_0 + \beta_1 x^2 + \mu$ is linear in parameters (we can define $z = x^2$ and estimate $y = \beta_0 + \beta_1 z + \mu$)
- $\log(y) = \beta_0 + \beta_1 x + \mu$ is linear in parameters
- $y = \beta_0 + \beta_1 \log(x) + \mu$ is linear in parameters

What would violate this assumption: Models where parameters enter non-linearly, such as $y = \beta_0 + x^{\beta_1} + \mu$ or $y = \frac{1}{1+e^{-\beta_0 - \beta_1 x}} + \mu$ (a logistic function). These require different estimation techniques.

3.7.2 Assumption 2: Random Sampling

> **Gauss-Markov Assumption 2: Random Sampling**
>
> We have a random sample of size n from the population: $\{(x_i, y_i) : i = 1, ..., n\}$

This assumption states that each observation in our dataset is drawn randomly and independently from the population of interest. Each unit in the population has an equal chance of being selected, and the selection of one unit doesn't affect the probability of selecting another.

Why it matters: Random sampling ensures that our sample is representative of the population. If certain types of observations are more likely to be included, our estimates may not generalize to the full population.

When this might be violated:

- **Self-selection:** If people choose whether to participate in a survey, those who respond may differ systematically from those who don't. For example, a wage survey might oversample high earners if low-wage workers are less likely to respond.

- **Convenience sampling:** Surveying only people who are easy to reach (e.g., college students for a study about the general population).
- **Time series data:** Observations collected over time are often correlated with each other. Today's stock price depends on yesterday's stock price, violating independence.
- **Clustered data:** Students within the same classroom, or workers within the same firm, may be more similar to each other than to randomly selected individuals.

3.7.3 Assumption 3: Variation in the Independent Variable

> **Gauss-Markov Assumption 3: Sample Variation in x**
>
> The independent variable has some variation in the sample: $\sum(x_i - \bar{x})^2 > 0$

This assumption simply requires that x is not constant in our sample. If everyone in our sample has the same value of x, we cannot estimate how changes in x relate to changes in y.

Why it matters: Look back at our formula for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

If all x_i equal $\bar{x}$, the denominator is zero, and the formula is undefined. Intuitively, if everyone has 12 years of education, we have no way to estimate what would happen with 13 years of education.

In practice: This assumption is almost always satisfied. It would only fail in unusual circumstances, like if you accidentally collected a sample where everyone happened to have the exact same value of x. However, having *little* variation in x (even if not zero) can lead to imprecise estimates, as we'll see when discussing variance.

3.7.4 Assumption 4: Zero Conditional Mean

> **Gauss-Markov Assumption 4: Zero Conditional Mean**
>
> The expected value of the error term is zero for any value of x: $E[\mu|x] = 0$

This is the most important and most frequently violated assumption. It states that the average value of all the unobserved factors affecting y is the same regardless of the value of x.

Understanding the assumption: Let's break this down carefully. The error term μ contains everything that affects y other than x. The zero conditional mean assumption says that if we could collect all people with $x = 10$ and compute the average of their μ values, it would be zero. And if we collected all people with $x = 20$ and averaged their μ values, that would also be zero. The average of unobservables is the same at every level of x.

Example: Education and Wages

Consider the regression:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

The error term μ includes ability, motivation, family connections, and other unobserved factors. The zero conditional mean assumption requires:

$$E[\mu|educ = 8] = E[\mu|educ = 12] = E[\mu|educ = 16] = 0$$

Is this plausible? Probably not. People with higher ability tend to get more education (because school is easier for them, or because they enjoy learning). So among people with 16 years of education, average ability is probably higher than among people with 8 years. This means $E[\mu|educ = 16] > E[\mu|educ = 8]$, violating the assumption.

Why it matters: When this assumption fails, OLS is **biased**. The estimate $\hat{\beta}_1$ will systematically overshoot or undershoot the true β_1. In the education example, if high-ability people get more education, our estimate of the "return to education" will partly capture the "return to ability," making education look more valuable than it really is.

This is the essence of **omitted variable bias**, which we'll explore in detail shortly.

3.7.5 Assumption 5: Homoskedasticity

> **i** Gauss-Markov Assumption 5: Homoskedasticity
>
> The variance of the error term is constant across all values of x: $Var(\mu|x) = \sigma^2$

This assumption states that the spread of the error term is the same regardless of the value of x. The "noise" in our data is equally noisy everywhere.

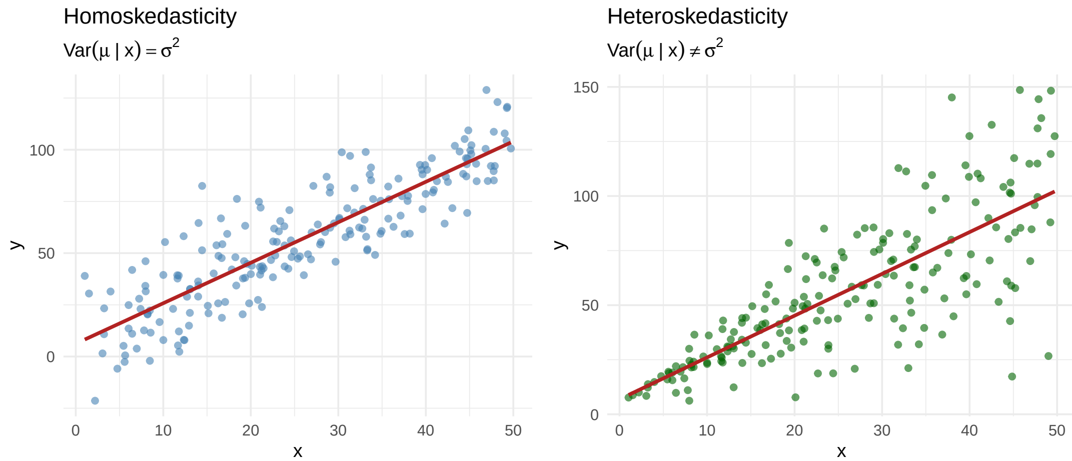
Understanding the assumption: While Assumption 4 concerns the *mean* of μ at different values of x, Assumption 5 concerns the *variance*. Homoskedasticity (from Greek: "same scatter") means the scatter of points around the regression line is constant. **Heteroskedasticity** (different scatter) occurs when the variance of μ depends on x.

Example: Income and Spending

Consider a regression of consumption spending on income. For low-income households, spending is relatively constrained—there's not much room for variation because most income goes to necessities. For high-income households, there's much more discretion in spending patterns, leading to greater variation. This would create heteroskedasticity: $Var(\mu|income)$ increases with income.

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```

Figure 3.3: Left: Homoskedasticity—the spread of points around the line is constant. Right: Heteroskedasticity—the spread increases with x.



Why it matters: Unlike Assumption 4, violating homoskedasticity does **not** make OLS biased. Your estimate of β_1 is still centered on the true value. However, heteroskedasticity affects the *variance* and *standard errors* of the estimator. The usual standard error formulas become incorrect, which can lead to invalid hypothesis tests and confidence intervals.

We'll discuss how to handle heteroskedasticity later in the course using **robust standard errors**.

3.7.6 Summary: Which Assumptions Matter for What?

Assumption	Needed for Unbiasedness?	Needed for Valid Standard Errors?
1. Linear in parameters	Yes	Yes
2. Random sampling	Yes	Yes

Assumption	Needed for Unbiasedness?	Needed for Valid Standard Errors?
3. Variation in x	Yes (otherwise undefined)	Yes
4. Zero conditional mean	**Yes**	Yes
5. Homoskedasticity	No	**Yes**

The key insight: Assumptions 1-4 are required for OLS to be **unbiased** (on average, we hit the target). Assumption 5 is additionally required for the usual standard error formulas to be valid and for OLS to be **BLUE** (Best Linear Unbiased Estimator).

3.8 Unbiasedness of OLS

Before discussing unbiasedness, let's revisit an important distinction we introduced in Appendix B: the difference between the **population** and the **sample**.

3.8.1 Population Parameters vs. Sample Estimates

The **population** represents the entire group we're interested in studying. In our wage-education example, the population might be all workers in the U.S. economy. The population regression model describes the *true* relationship in this population:

$$y = \beta_0 + \beta_1 x + \mu$$

The parameters β_0 and β_1 are **population parameters**—fixed, unknown constants that describe how x and y are related for everyone in the population. We never observe these true values directly.

Instead, we collect a **sample**—a subset of observations from the population. Using this sample, we compute **estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$. These estimates are our best guesses for the unknown population parameters.

Here's the key insight: *if we drew a different sample, we would get different estimates.* Your classmate analyzing a different random sample of workers would compute different values of $\hat{\beta}_0$ and $\hat{\beta}_1$ than you would. This randomness in the estimates is called **sampling variability**.

3.8.2 What Does Unbiasedness Mean?

Given that estimates vary from sample to sample, how do we know if our estimation procedure is any good? One desirable property is **unbiasedness**.

An estimator $\hat{\theta}$ is **unbiased** if its expected value equals the true population parameter:

$$E[\hat{\theta}] = \theta$$

Unbiasedness does *not* mean that any particular estimate exactly equals the truth. Rather, it means that if we could:

1. Draw a random sample from the population
2. Compute $\hat{\beta}_1$ from that sample
3. Repeat steps 1-2 many, many times
4. Average all the $\hat{\beta}_1$ values we computed

…then that average would equal the true β_1.

In other words, our estimation procedure doesn't systematically overshoot or undershoot the target. Individual estimates might be too high or too low, but on average, we get it right.

3.8.3 Proof Sketch

Starting from our formula for $\hat{\beta}_1$ and substituting the true model $y_i = \beta_0 + \beta_1 x_i + \mu_i$:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})\mu_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Taking expectations and using the zero conditional mean assumption ($E[\mu_i|x_i] = 0$):

$$E[\hat{\beta}_1] = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x}) \cdot 0}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \beta_1$$

Thus, OLS is unbiased when the Gauss-Markov assumptions hold.

3.8.4 Simulation: Unbiasedness in Action

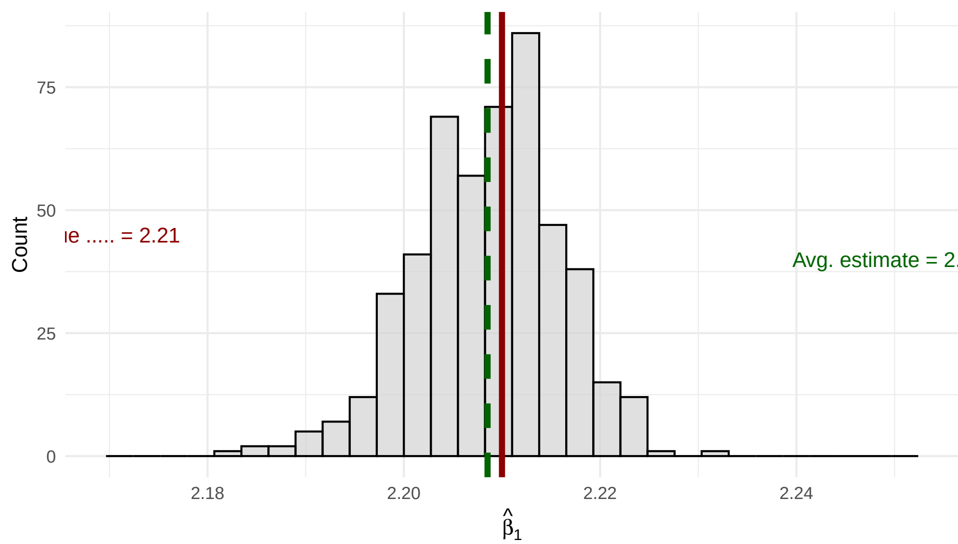
Show Code: Unbiasedness Simulation

```r
# Create a "population" with known parameters
set.seed(1248)
n_pop <- 1000
x_pop <- sample(1:100, size = n_pop, replace = TRUE)
err_pop <- rnorm(n_pop)

# True model: y = 3.5 + 2.21*x + error
pop_data <- tibble(
  x = x_pop,
  y = 3.5 + 2.21 * x + err_pop
)

# Draw many samples and estimate beta_1 each time
set.seed(812476)
n_simulations <- 500
sample_size <- 25

sim_results <- map_dfr(1:n_simulations, function(i) {
  # Draw a random sample
  sample_data <- slice_sample(pop_data, n = sample_size, replace = TRUE)

  # Estimate OLS
  reg <- lm(y ~ x, data = sample_data)

  tibble(
    iteration = i,
    b1_hat = coef(reg)[2]
  )
})

# Calculate the average of all estimates
mean_b1 <- mean(sim_results$b1_hat)
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<ce>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<b2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<82>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<81>
```

Figure 3.4: Distribution of OLS slope estimates across 500 samples. The green line shows the average of all estimates, which equals the true value (2.21), demonstrating unbiasedness.



The simulation in Figure 3.4 confirms the theory: while individual estimates vary around the true value, the average across many samples equals the true parameter.

3.9 Omitted Variable Bias

What happens when the zero conditional mean assumption fails? The most common culprit is **omitted variable bias**: we've left out a variable that affects both x and y.

Consider the true model:

$$y = \beta_0 + \beta_1 x + \beta_2 z + \mu$$

If we omit z and estimate only $y = \beta_0 + \beta_1 x + \nu$ (where $\nu = \beta_2 z + \mu$), then whenever z is correlated with x, we have $E[\nu|x] \neq 0$, and our estimate of β_1 is biased.

Example: In the education-wage regression, if ability (z) affects both education choices (x) and wages (y), then omitting ability creates bias. The estimated "return to education" will partly reflect the return to ability.

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<ce>
```

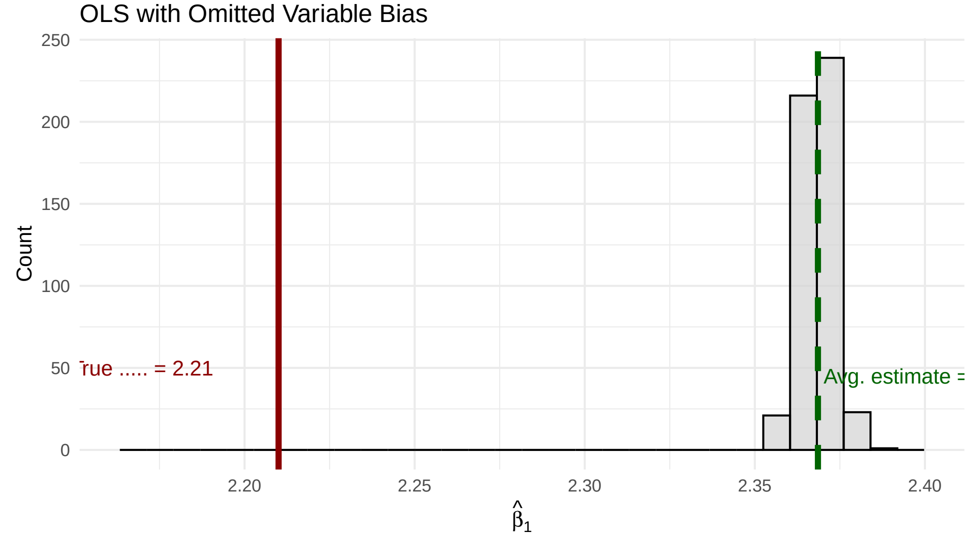
```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<b2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<e2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<82>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<81>
```

Figure 3.5: With omitted variable bias, the average of OLS estimates (green) systematically differs from the true value (red). The bias persists even with many samples.



3.10 Variance of the OLS Estimator

Even when OLS is unbiased, our estimates vary from sample to sample. The **variance** of $\hat{\beta}_1$ tells us how much variation to expect.

Under the Gauss-Markov assumptions:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x}$$

where $\sigma^2 = Var(\mu)$ is the error variance and $SST_x = \sum(x_i - \bar{x})^2$ is the total variation in x.

This formula reveals two key insights:

1. **More noise (σ^2) means more variance:** If there's a lot of unexplained variation in y, our estimates are less precise.
2. **More variation in x means less variance:** Spreading out our observations of x gives us more information and more precise estimates. This also implies that larger samples (which have more total variation) yield more precise estimates.

3.11 Standard Errors

The **standard error** of $\hat{\beta}_1$ is the square root of the variance:

$$se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SST_x}}$$

where $\hat{\sigma} = \sqrt{\frac{SSR}{n-2}}$ is the estimated standard deviation of the errors.

Standard errors quantify the precision of our estimates and are essential for hypothesis testing and confidence intervals (which we'll cover in the next chapter).

```r
# R reports standard errors automatically
summary(lm_mpg)
```

```

Call:
lm(formula = mpg ~ hp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.7121 -2.1122 -0.8854  1.5819  8.2360 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
hp          -0.06823    0.01012  -6.742 1.79e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared:  0.6024,	Adjusted R-squared:  0.5892 
F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

The standard error of $\hat{\beta}_1$ is 0.01, shown in the `Std. Error` column.

3.12 OLS is BLUE

Under all five Gauss-Markov assumptions, OLS is **BLUE**:

- **B**est: Lowest variance among all linear unbiased estimators
- **L**inear: The estimator is a linear function of the y_i's
- **U**nbiased: $E[\hat{\beta}] = \beta$
- **E**stimator: It's a rule for using sample data to estimate population parameters

This is a powerful result: among all estimators that are linear and unbiased, OLS has the smallest variance. You cannot do better without either using non-linear methods or accepting some bias.

3.13 Summary and Conclusion

This chapter introduced Ordinary Least Squares regression, the foundational tool of econometrics. We started with the goal of modeling how the conditional expectation $E[Y|X]$ varies with x, and showed how the linear regression model $y = \beta_0 + \beta_1 x + \mu$ provides a framework for this.

We derived the OLS estimator formulas, learned to compute regressions in R, and explored the properties of the OLS regression line. We then turned to the statistical properties of OLS, introducing the Gauss-Markov assumptions and proving that OLS is unbiased when these assumptions hold. When the zero conditional mean assumption fails—typically due to omitted variables—our estimates become biased.

Finally, we discussed the variance of OLS estimates and the conditions under which OLS is the best linear unbiased estimator.

3.13.1 Key Takeaways

- **The goal** is to estimate the conditional expectation $E[Y|X] = \beta_0 + \beta_1 x$
- **The OLS slope** equals the covariance of x and y divided by the variance of x:

$$\hat{\beta}_1 = \frac{Cov(x, y)}{Var(x)}$$

- **Bivariate OLS is essentially correlation**—be very careful about causal interpretation
- **The Gauss-Markov assumptions** (especially zero conditional mean) are required for unbiasedness
- **Omitted variable bias** arises when we exclude a variable correlated with both x and y
- **OLS is BLUE** under the Gauss-Markov assumptions—best among linear unbiased estimators

3.14 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

References

Angrist, Joshua D., and Jörn-Steffen Pischke. 2014. *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press.

Bailey, Michael A. 2020. *Real Econometrics: The Right Tools to Answer Important Questions*. Oxford University Press.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media, Incorporated.

Wooldridge, Jeffrey M. 2019. *Introductory Econometrics: A Modern Approach*. 7th ed. Cengage Learning.

A Intro to R and RStudio

Key Questions

- How do I install R and RStudio on my computer?
- What are the main components of the RStudio interface?
- How do I write and run R code?
- What are objects and how do I use them?
- How do I install and load R packages?

Suggested Readings

- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1-3

This appendix will walk you through installing R and RStudio, understanding the RStudio interface, and writing your first lines of R code. By the end of this chapter, you will be ready to start working with data and running econometric analyses.

A.1 Installing R

To get started with R, you first need to download and install the base R software. R is available for free from CRAN (the Comprehensive R Archive Network). Navigate to https://cloud.r-project.org/ and click the download link for your operating system (Windows, macOS, or Linux). Follow the installation instructions provided for your system.

Once the installation is complete, you will have R installed on your computer. However, we will not typically interact with R directly. Instead, we will use RStudio, an integrated development environment (IDE) that makes working with R much more convenient.

A.2 Installing RStudio

RStudio is a powerful IDE designed specifically for working with R. It provides a user-friendly interface with helpful features like syntax highlighting, code completion, and integrated help documentation. You can download RStudio Desktop for free from https://posit.co/download/

rstudio-desktop/. The website should automatically detect your operating system and show you the appropriate download link. Click the link and follow the installation instructions.

> **!** Only Use RStudio
>
> After installing both R and RStudio, you will only ever need to open RStudio. RStudio runs R in the background, so you don't need to open the base R application separately. Feel free to delete the shortcut for base R from your desktop if you wish. When you open an R script file (`.R`) for the first time, be sure to set RStudio as the default program to open such files.

A.3 The RStudio Interface

When you first open RStudio, you will see an interface divided into several panels (or "panes"), as seen in Figure A.1. Each panel serves a different purpose in your workflow. Understanding these panels is essential for working efficiently in R.

A.3.1 The Script Editor

The Script Editor is where you write your R code in script files. A script file is simply a text file containing R code that can be saved, edited, and run again later. Working in script files is essential for reproducibility—you can always go back and see exactly what you did, make changes, and re-run your analysis. To open a new script file, go to File → New File → R Script, or use the keyboard shortcut Cmd+Shift+N (Mac) or Ctrl+Shift+N (Windows/Linux).

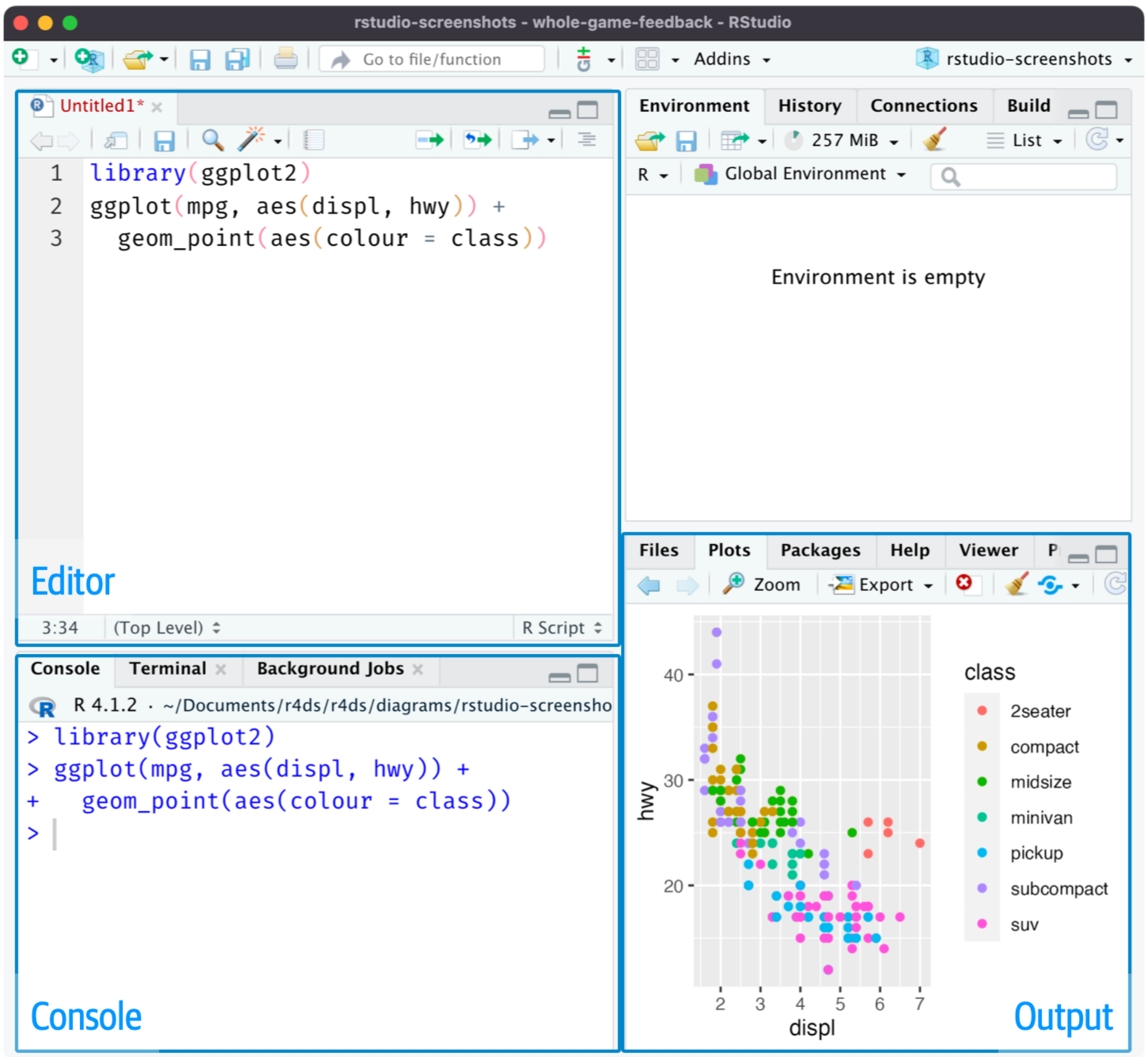
A.3.2 The Console

The Console is where R actually executes your code. When you run code from a script, it gets sent to the console line by line. You can also type code directly into the console and press Enter to run it immediately. However, code typed directly into the console is not saved, so this approach is generally only recommended for quick tests or exploratory work. Output from many functions, including regression results, will be printed to the console.

A.3.3 The Environment Panel

The Environment panel displays all the "objects" you have created in your current R session. These objects might include data frames, vectors, or stored values. You can see each object's name along with a brief summary of its contents. This panel is useful for keeping track of what data and variables you currently have available.

Figure A.1: The RStudio interface with its four main panels labeled.



A.3.4 The Plots and Output Panel

This panel serves multiple purposes. When you create visualizations (like scatter plots or histograms), they will appear here. This panel also contains tabs for browsing files on your computer, viewing installed packages, and accessing R's built-in help documentation.

> Customizing Your Layout
>
> You can hide or show different panels under the "View" menu. This is useful when you want more screen space for a particular task, such as focusing on your script while writing code.

A.4 Basic Coding in R

Now that RStudio is set up, let's write some code! We'll start with simple operations in the console to get familiar with R's syntax.

A.4.1 Arithmetic Operations

R can function as a sophisticated calculator. Try typing the following expressions in the console and pressing Enter:

```r
2 + 2
```

```
[1] 4
```

```r
20 * 5 / 87
```

```
[1] 1.149425
```

```r
sin(pi / 2)
```

```
[1] 1
```

```r
sqrt(6)
```

```
[1] 2.44949
```

As you can see, R can handle basic arithmetic as well as mathematical functions like `sin()` and `sqrt()`. The `pi` you see in the third example is a built-in constant in R representing the mathematical constant π.

A.4.2 Objects and Assignment

R is an "object-oriented language," meaning that data, variables, and nearly everything else is stored as an "object." You create objects using the assignment operator `<-`. The assignment operator takes whatever is on the right side and stores it in the name on the left side.

```
x <- 2 + 5
```

After running this code, the value `7` is now stored in an object called `x`. Notice that R didn't print anything—it just stored the value. To see what's stored in an object, type its name and press Enter:

```
x
```

```
[1] 7
```

You can use objects in subsequent calculations, just like variables in algebra:

```
x + 10
```

```
[1] 17
```

You can also update an object by reassigning it:

```
x <- x + 50
x
```

```
[1] 57
```

> **Naming Objects**
>
> When creating object names, keep them short but descriptive. Object names cannot contain spaces. Some good examples include: `flight_data`, `reg_controls_1`, `acs_male`, or `X_mat`.

A.4.3 Vectors

A vector is the simplest type of object in R. Vectors contain a sequence of values of the same type (all numbers, all text, etc.). You create vectors using the `c()` function, which stands for "combine":

```
primes <- c(2, 3, 5, 7, 11, 13)
primes
```

```
[1]  2  3  5  7 11 13
```

One powerful feature of R is that arithmetic operations on vectors are applied element-by-element:

```
primes * 2
```

```
[1]  4  6 10 14 22 26
```

```
primes - 1
```

```
[1]  1  2  4  6 10 12
```

You can also perform arithmetic between two vectors of the same length:

```
odds <- c(1, 3, 5, 7, 9, 11)
primes + odds
```

```
[1]  3  6 10 14 20 24
```

A.4.4 Functions

R has a large collection of built-in functions that perform specific tasks. Functions are called using the syntax: `function_name(argument1 = value1, argument2 = value2, ...)`.

For example, the `seq()` function generates a sequence of numbers:

```
bb <- seq(from = 1, to = 30, by = 3)
bb
```

```
 [1]  1  4  7 10 13 16 19 22 25 28
```

Here, `from`, `to`, and `by` are the *arguments* of the function. We set `from = 1` to start at 1, `to = 30` to end at 30, and `by = 3` to increment by 3.

> Getting Help
>
> To learn more about any function, type `?` followed by the function name in the console. For example, `?seq` will open the documentation for the `seq()` function.

A.4.5 Data Frames

Data frames are the primary way that structured, tabular data is stored in R. A data frame is essentially a table where each column is a vector and each row is an observation. R comes with several built-in data frames for practice, including `mtcars`, which contains data on various car models.

Use `head()` to view the first few rows of a data frame:

```r
head(mtcars)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

You can explore the structure of a data frame using several helpful functions:

```r
nrow(mtcars)
```

```
[1] 32
```

```r
ncol(mtcars)
```

```
[1] 11
```

```r
colnames(mtcars)
```

```
 [1] "mpg"  "cyl"  "disp" "hp"   "drat" "wt"   "qsec" "vs"   "am"   "gear"
[11] "carb"
```

To access a specific column (variable) within a data frame, use the `$` operator:

```r
head(mtcars$mpg)
```

```
[1] 21.0 21.0 22.8 21.4 18.7 18.1
```

Here, `mtcars$mpg` extracts the `mpg` column from the `mtcars` data frame as a vector. I wrapped it in `head()` to show only the first few values.

A.4.6 Comments

In R, anything preceded by the `#` symbol is treated as a comment and will not be executed. Comments are essential for documenting your code—explaining what each section does, leaving notes for yourself, or marking areas that need revision.

```r
# Find the mean miles per gallon
mean(mtcars$mpg)
```

```
[1] 20.09062
```

Good commenting habits make your code readable to others (and to your future self).

A.5 Working with R Scripts

While the console is useful for quick tests, you should do the vast majority of your work in script files. Scripts allow you to save your code, edit it easily, and reproduce your entire analysis by running the script from top to bottom.

A.5.1 Creating and Running Scripts

To create a new script, go to File → New File → R Script (or use Cmd/Ctrl + Shift + N). Write your code in the script editor, then use the following keyboard shortcuts to run it:

Cmd/Ctrl + Enter executes the current line (or highlighted selection) in the console. The cursor then automatically moves to the next line, making it easy to step through your code line by line.

Figure A.2: Example R script with cursor position shown. Pressing Cmd/Ctrl + Enter will run the complete command that creates the `not_cancelled` object, then move the cursor to the next statement.

```
library(dplyr)
library(nycflights13)

not_cancelled <- flights |>
  filter(!is.na(dep_delay)█, !is.na(arr_delay))

not_cancelled |>
  group_by(year, month, day) |>
  summarize(mean = mean(dep_delay))
```

Cmd/Ctrl + Shift + S runs the entire script from top to bottom. This is useful for checking that your complete analysis works correctly and produces reproducible results.

A.5.2 Script Best Practices

Following good practices when writing scripts will save you time and frustration. Always start your script with your name and a brief description of the script's purpose in comments. Load all required packages at the beginning of the script using `library()` so anyone reading your code knows what dependencies are needed. Never include `install.packages()` in your script—you only need to install packages once on your computer, and you don't want your script to install packages on someone else's machine without their permission. Save your scripts with short, descriptive names that contain no spaces, such as `assignment_3.R` or `ec_325_data_plotting.R`.

A.6 Working Directories

R uses a concept called the "working directory"—the folder on your computer where R looks for files you want to load and where it saves files you create. You can see your current working directory at the top of the console, or by running:

```
getwd()
```

A good workflow is to save your R script in a dedicated folder for this course (or a subfolder for each week's work), put all related data files in the same folder, and then set your working directory to that location. In RStudio, go to Session → Set Working Directory → To Source File Location to automatically set the working directory to wherever your script is saved.

A.7 R Packages

R comes with a set of "base" packages that provide fundamental functionality. However, the true power of R lies in its vast ecosystem of user-contributed packages. These packages extend R's capabilities for specialized tasks—from data visualization to complex statistical modeling.

A.7.1 Installing Packages

You only need to install a package once on your computer. Use the `install.packages()` function:

```
install.packages("tidyverse")
```

A.7.2 The Tidyverse

The tidyverse is a collection of R packages designed for data science. These packages share a common philosophy and work seamlessly together, making data manipulation and visualization more intuitive. Some of the most commonly used tidyverse packages include `dplyr` (data manipulation), `ggplot2` (visualization), and `readr` (reading data files).

Installing `tidyverse` installs all of these packages at once.

A.7.3 Loading Packages

Once a package is installed, you need to load it each time you start a new R session using `library()`:

```
library(tidyverse)
```

When you load the tidyverse, you'll see a message listing which packages were attached. You may also see messages about "conflicts"—this just means some tidyverse functions have the same name as functions in base R, and the tidyverse versions will take precedence. This is normal and expected behavior.

> **!** Remember the Difference
>
> `install.packages()` downloads and installs a package to your computer (do this once). `library()` loads an already-installed package into your current R session (do this every time you start R).

A.8 Summary and Conclusion

This appendix walked you through the essential first steps for working with R. You installed R and RStudio, learned how the RStudio interface is organized, and wrote your first R code. You now understand how to create and manipulate objects, work with vectors and data frames, write and run scripts, and install packages.

A.8.1 Key Takeaways

- **RStudio is your interface to R:** Always work in RStudio, not base R. The script editor, console, environment, and plots panels each serve distinct purposes in your workflow.
- **Scripts are essential for reproducibility:** Write your code in script files that you can save, edit, and re-run. Use comments to document what your code does.
- **Objects store everything in R:** Use the assignment operator `<-` to create objects. Vectors hold sequences of values; data frames hold structured tabular data.
- **Packages extend R's capabilities:** Install packages once with `install.packages()`, then load them each session with `library()`. The tidyverse is a particularly useful collection of packages for data science.

- **Working directories matter:** Set your working directory to the folder containing your script and data files to keep your analysis organized.

With these foundations in place, you are ready to start loading real data, creating visualizations, and running econometric analyses.

A.9 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

B Statistics and Probability Review

Key Questions

- What are the properties of the summation operator?
- How do we interpret derivatives and partial derivatives?
- How do we find the maximum or minimum of a function?
- What is the difference between a population and a sample?
- What is a random variable and how do we describe its distribution?
- How do we measure central tendency, variability, and relationships between variables?
- What is the normal distribution and why is it important?

Suggested Readings

- Wooldridge (2019), Appendix A, B, and C

This appendix reviews the essential mathematical and statistical concepts that form the foundation for econometrics. If you've taken introductory statistics or calculus, much of this material will be familiar. However, econometrics uses these tools in specific ways, so it's worth reviewing them with an eye toward how they'll appear throughout the course.

B.1 Basic Mathematical Tools

B.1.1 The Summation Operator

The summation operator provides a compact way to write the sum of a sequence of numbers. If $\{x_i : i = 1, 2, ..., n\}$ is a sequence of n numbers, then the sum of these numbers is written as:

$$x_1 + x_2 + ... + x_n = \sum_{i=1}^{n} x_i$$

The symbol Σ (capital Greek letter sigma) tells us to add up the terms. The expression $i = 1$ below the sigma indicates that we start counting at $i = 1$, and the n above tells us to stop at $i = n$.

B.1.1.1 Properties of Summation

The summation operator has several useful properties that we will use frequently.

Property 1: Summing a constant. If we add a constant c to itself n times, we get n times the constant:

$$\sum_{i=1}^{n} c = nc$$

Property 2: Factoring out constants. We can pull constants outside the summation:

$$\sum_{i=1}^{n} cx_i = c\sum_{i=1}^{n} x_i$$

Property 3: Separating additive terms. Sums can be split across addition:

$$\sum_{i=1}^{n}(ax_i + by_i) = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} y_i$$

> ⚠ Common Mistakes with Summation
>
> These properties do **not** extend to division or multiplication in the ways you might expect. We **cannot** break up divided expressions:
>
> $$\sum_{i=1}^{n} \frac{x_i}{y_i} \neq \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}$$
>
> We **cannot** break up multiplicative expressions:
>
> $$\sum_{i=1}^{n} x_i^2 \neq \left(\sum_{i=1}^{n} x_i\right)^2$$

B.1.1.2 The Sample Mean

Given n numbers $\{x_i : i = 1, 2, ..., n\}$, we can compute their **average** or **mean** value by adding them up and dividing by n. The average value, denoted $\bar{x}$ (read "x-bar"), is written as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

The average is an example of a **descriptive statistic**: a value that describes or summarizes a characteristic of a set of data points. In this case, the mean describes the **central tendency** of x_i—where the "middle" of the data lies.

B.1.1.3 Useful Summation Results

Two results involving summations will appear repeatedly in this course.

Result 1: Deviations from the mean sum to zero. The sum of the deviations $(x_i - \bar{x})$ from the mean always equals zero:

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

This can be proven using the properties above:

$$\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x} = \sum_{i=1}^{n} x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0$$

The last step follows because $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ implies $n\bar{x} = \sum_{i=1}^{n} x_i$.

Result 2: Sum of squared deviations. The sum of squared deviations can be rewritten as:

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

More generally, for two variables x and y:

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$

These identities are useful for computing variances and covariances, which we will discuss shortly.

B.1.2 Derivatives

The derivative $\frac{dy}{dx}$ of a function $y = f(x)$ measures its **instantaneous rate of change**. It tells you how sensitive the function's output is to a small change in its input. In econometrics, derivatives help us interpret how changes in independent variables affect dependent variables.

Here are derivatives of some common functional forms in econometrics:

Function	Derivative
$y = \beta_0 + \beta_1 x$	$\frac{dy}{dx} = \beta_1$
$y = \beta_0 + \beta_1 x + \beta_2 x^2$	$\frac{dy}{dx} = \beta_1 + 2\beta_2 x$
$y = \beta_0 + \frac{\beta_1}{x}$	$\frac{dy}{dx} = -\frac{\beta_1}{x^2}$
$y = \beta_0 + \beta_1 \ln(x)$	$\frac{dy}{dx} = \frac{\beta_1}{x}$

Notice that in the linear case, the derivative is simply the coefficient β_1—the effect of x on y is constant. In the other cases, the derivative depends on the value of x, meaning the effect of x on y varies depending on where you start.

B.1.3 Partial Derivatives

When y is a function of multiple variables, say $y = f(x_1, x_2)$, we use **partial derivatives** to measure how y changes with respect to one variable while holding the others constant.

The partial derivative of y with respect to x_1, holding x_2 constant, is denoted $\frac{\partial y}{\partial x_1}$. Similarly, the partial derivative with respect to x_2, holding x_1 constant, is $\frac{\partial y}{\partial x_2}$.

For example, if $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, then:

$$\frac{\partial y}{\partial x_1} = \beta_1, \qquad \frac{\partial y}{\partial x_2} = \beta_2$$

The concept of "holding other variables constant" is central to econometrics. When we estimate a regression coefficient, we are trying to measure the partial effect of one variable on the outcome, controlling for other factors.

B.1.4 Optimization: Maxima and Minima

In econometrics, we often need to find the value of a variable that maximizes or minimizes a function. For example, Ordinary Least Squares (OLS) regression finds coefficient estimates by minimizing the sum of squared residuals. Understanding optimization is essential for understanding how these estimates are derived.

B.1.4.1 Finding Critical Points

To find the maximum or minimum of a function $f(x)$, we take the derivative, set it equal to zero, and solve for x. The solutions are called **critical points**.

For example, consider the quadratic function:

$$f(x) = ax^2 + bx + c$$

Taking the derivative and setting it equal to zero:

$$\frac{df}{dx} = 2ax + b = 0$$

Solving for x:

$$x^* = -\frac{b}{2a}$$

This critical point x^* is where the function reaches its maximum or minimum.

B.1.4.2 Determining Maximum vs. Minimum

How do we know if a critical point is a maximum or a minimum? We use the **second derivative test**. The second derivative tells us about the curvature of the function:

$$\frac{d^2f}{dx^2} = 2a$$

The rule is:

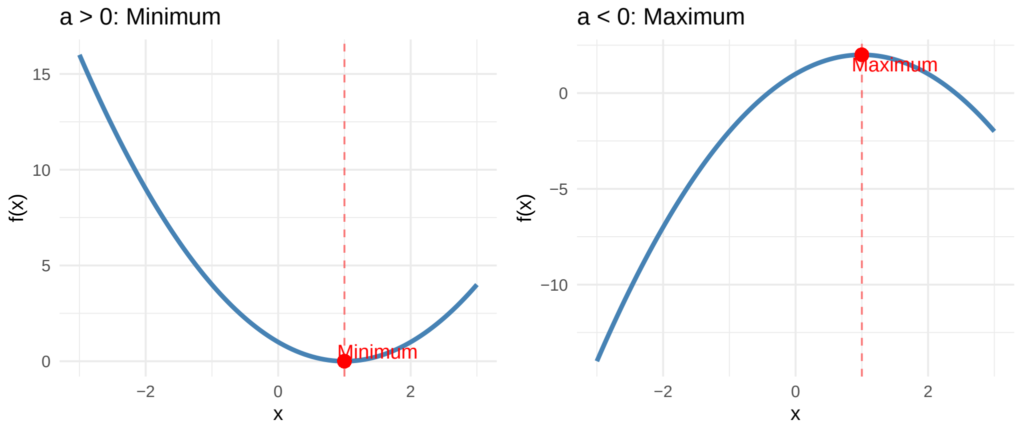
- If $\frac{d^2f}{dx^2} > 0$ at the critical point, the function is **concave up** (bowl-shaped), and we have a **minimum**
- If $\frac{d^2f}{dx^2} < 0$ at the critical point, the function is **concave down** (hill-shaped), and we have a **maximum**

For our quadratic function $f(x) = ax^2 + bx + c$:

- If $a > 0$: The parabola opens upward, and $x^* = -\frac{b}{2a}$ is a **minimum**
- If $a < 0$: The parabola opens downward, and $x^* = -\frac{b}{2a}$ is a **maximum**

```
Warning in geom_point(aes(x = 1, y = 0), color = "red", size = 3): All aesthetics have lengt
i Please consider using `annotate()` or provide this layer with data containing
  a single row.

Warning in geom_point(aes(x = 1, y = 2), color = "red", size = 3): All aesthetics have lengt
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

Figure B.1: Left: When a > 0, the parabola opens upward and the critical point is a minimum. Right: When a < 0, the parabola opens downward and the critical point is a maximum.



Optimization in OLS

In OLS regression, we minimize the sum of squared residuals, which is a quadratic function of the coefficients. The second derivative is positive (the function is convex), confirming that the solution is indeed a minimum. This is why setting the first derivative equal to zero gives us the best-fitting line.

B.1.4.3 Optimization with Multiple Variables

When a function depends on multiple variables, we find critical points by setting all partial derivatives equal to zero simultaneously. For a function $f(x_1, x_2)$:

$$\frac{\partial f}{\partial x_1} = 0 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 0$$

Determining whether a critical point is a maximum, minimum, or neither (a saddle point) requires examining the second partial derivatives, which we will encounter when deriving the OLS estimator.

B.2 Population vs. Sample

Before diving into probability, it's important to understand the distinction between a **population** and a **sample**, as this distinction underlies all of statistical inference.

The **population** is the entire group of individuals, objects, or observations that we are interested in studying. For example, if we want to understand the wages of all U.S. workers, the population is every person employed in the United States. If we want to know the effect of a drug on blood pressure, the population might be all adults with hypertension.

In practice, we almost never observe the entire population. Instead, we collect data on a **sample**—a subset of the population. We might survey 5,000 workers rather than all 160 million, or run a clinical trial with 500 patients rather than everyone with hypertension.

> **!** The Goal of Statistical Inference
>
> The fundamental goal of statistics is to use information from a **sample** to draw conclusions about the **population**. We observe sample statistics (like the sample mean $\bar{x}$) and use them to make inferences about population parameters (like the population mean μ).

This distinction affects our notation:

Concept	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Correlation	ρ	r

Population parameters are typically denoted with Greek letters and are usually unknown. Sample statistics are computed from our data and are used to estimate the population parameters.

For example, if we want to know the average wage of all U.S. workers (μ), we collect a sample and compute the sample mean ($\bar{x}$). Under certain conditions, $\bar{x}$ is a good estimate of μ. Much of econometrics is about understanding when and how well sample statistics estimate population parameters, and quantifying the uncertainty in those estimates.

B.3 Probability Basics

B.3.1 Random Variables

A **random variable** is a variable whose value is determined by a random phenomenon. It represents a number, but the exact value is determined by chance. Examples include the outcome of a dice roll, whether a coin lands heads or tails, or the percentage of flights that arrive on time.

Most variables we work with in econometrics can be thought of as random variables. A person's wage, years of education, or health status all have elements of randomness—we cannot perfectly predict them in advance.

B.3.2 Probability

Probability quantifies uncertainty. If X is a random variable that takes on k possible values $\{x_1, x_2, ..., x_k\}$, then the probability that X takes a particular value x_j is denoted:

$$p_j = P(X = x_j)$$

Each probability p_j must be between 0 and 1, and the probabilities must sum to 1: $p_1 + p_2 + ... + p_k = 1$.

B.3.3 Probability Distributions

The **probability density function (pdf)**, denoted $f(x)$, gives the probability that a random variable takes on a particular value x. For discrete random variables, $f(x) = P(X = x)$.

The **cumulative distribution function (cdf)**, denoted $F(x)$, gives the probability that the random variable is less than or equal to x:

$$F(x) = P(X \leq x)$$

The cdf is useful for answering questions like "what is the probability that X is below some threshold?"

B.3.4 Expected Values

The **expected value** (or **mathematical expectation**) of a random variable X, denoted $E[X]$, represents the "average" outcome we would expect if we could repeat the random process many times. It is our population "best guess" for the value of X. We often denote it with μ or μ_X.

For a discrete random variable with pdf $f(x)$, the expected value is the weighted average of all possible values, where the weights are the probabilities:

$$E[X] = \sum_{j=1}^{k} x_j f(x_j) = \mu$$

Example: Let X be the outcome of rolling a fair six-sided die. Each value (1 through 6) has probability 1/6. The expected value is:

$$E[X] = \frac{1}{6}(1) + \frac{1}{6}(2) + \frac{1}{6}(3) + \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6) = 3.5$$

Notice that 3.5 is not a possible outcome of any single roll, but it is the average we would expect over many rolls.

B.3.4.1 Properties of Expected Values

The properties of expected values mirror those of the summation operator:

Property 1: For any constant c, $E[c] = c$.

Property 2: For any constants a and b, $E[aX + b] = aE[X] + b$.

Property 3: If $\{a_1, a_2, ..., a_n\}$ are constants and $\{X_1, X_2, ..., X_n\}$ are random variables, then:

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

B.4 Measures of Variability

Knowing the central tendency of a variable is useful, but it doesn't tell us everything. Two distributions can have the same mean but look very different if one is tightly clustered around the mean while the other is spread out. We need measures of **variability** or **dispersion**.

B.4.1 Variance

Let X be a random variable with mean $\mu = E[X]$. The **variance** of X, denoted σ^2 or $\text{Var}(X)$, measures how spread out the distribution is:

$$\text{Var}(X) = E[(X - \mu)^2] = \sigma^2$$

The variance is the expected squared deviation from the mean. Squaring serves two purposes: it ensures all deviations are positive (so positive and negative deviations don't cancel out), and it gives more weight to observations far from the mean.

B.4.2 Standard Deviation

The **standard deviation** of X, denoted $\text{sd}(X)$ or σ, is the positive square root of the variance:

$$\text{sd}(X) = \sqrt{\text{Var}(X)} = \sigma$$

The standard deviation is often more interpretable than the variance because it is measured in the same units as X itself. If X is measured in dollars, the standard deviation is also in dollars, whereas the variance would be in "dollars squared."

B.4.3 Covariance

A central concern in econometrics is understanding how variables move together. When X is large, is Y also large? Or is it small? Or is there no relationship at all?

The **covariance** quantifies the linear relationship between two random variables. Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance is defined as:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

The covariance measures how X and Y move together:

- **Positive covariance:** When X is above its mean, Y tends to be above its mean (and vice versa). The variables move in the same direction.
- **Negative covariance:** When X is above its mean, Y tends to be below its mean. The variables move in opposite directions.
- **Zero covariance:** There is no linear relationship between X and Y.

B.4.4 Correlation Coefficient

The covariance tells us the direction of the relationship, but its magnitude depends on the scales of X and Y, making it hard to interpret. The **correlation coefficient** standardizes the covariance to produce a unitless measure:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \cdot \text{sd}(Y)} = \rho_{XY}$$

The correlation coefficient is always between -1 and 1:

- $\rho_{XY} = 1$: Perfect positive linear relationship
- $\rho_{XY} = -1$: Perfect negative linear relationship
- $\rho_{XY} = 0$: No linear relationship

Perfect correlations rarely occur in practice. Values of ρ_{XY} closer to 1 or -1 indicate stronger linear relationships.

> **i** Correlation vs. Causation
>
> Remember from Chapter 1: correlation does not imply causation! A strong correlation between X and Y tells us the variables move together, but it does not tell us whether X causes Y, Y causes X, or some third factor causes both.

B.4.5 Conditional Expectations

While correlation measures the overall linear relationship between two variables, we often want to describe how the expected value of one variable changes with another. This is where **conditional expectations** come in.

Let X and Y be random variables. The conditional expectation of Y given that X takes a specific value x is denoted:

$$E[Y|X = x]$$

This asks: what is our "best guess" for Y when we know that X equals x?

Example: What is the expected wage (Y) for people with a college degree ($X = 1$) versus those without ($X = 0$)? $E[Y|X = 1]$ and $E[Y|X = 0]$ answer these questions.

Conditional expectations are fundamental to regression analysis. When we estimate a regression of Y on X, we are essentially modeling how $E[Y|X]$ changes with X.

B.5 Distributions

B.5.1 The Normal Distribution

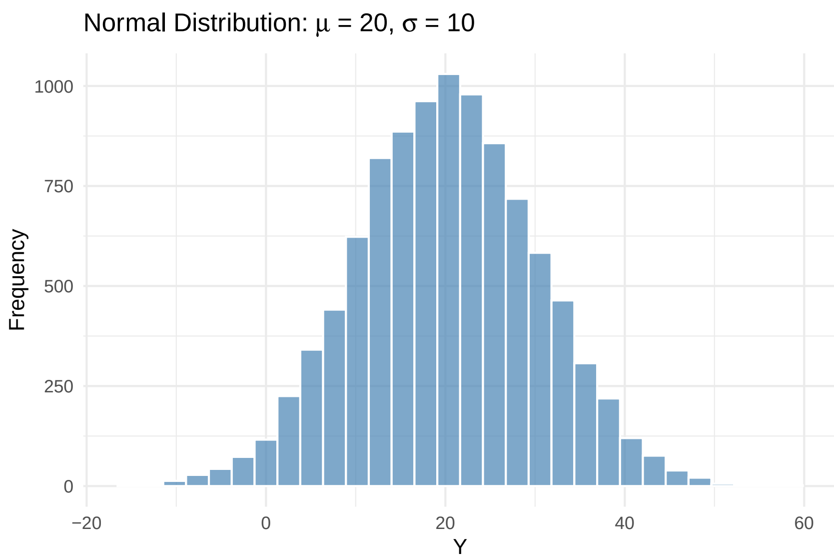
The **normal distribution** (also called the Gaussian distribution) is the most important distribution in statistics and econometrics. Many naturally occurring phenomena are approximately normally distributed, and many statistical procedures assume normality.

A random variable X with mean μ and variance σ^2 that follows a normal distribution is denoted:

$$X \sim N(\mu, \sigma^2)$$

The normal distribution has a characteristic "bell curve" shape: symmetric around the mean, with most observations clustered near the center and fewer observations in the tails.

Figure B.2: A histogram of 10,000 draws from a normal distribution with mean 20 and standard deviation 10.



Two key facts about the normal distribution:

- Approximately **68%** of observations lie within one standard deviation of the mean (between $\mu - \sigma$ and $\mu + \sigma$)
- Approximately **95%** of observations lie within two standard deviations of the mean (between $\mu - 2\sigma$ and $\mu + 2\sigma$)

B.5.2 The Standard Normal Distribution

The **standard normal distribution** is a special case of the normal distribution with mean 0 and variance 1:

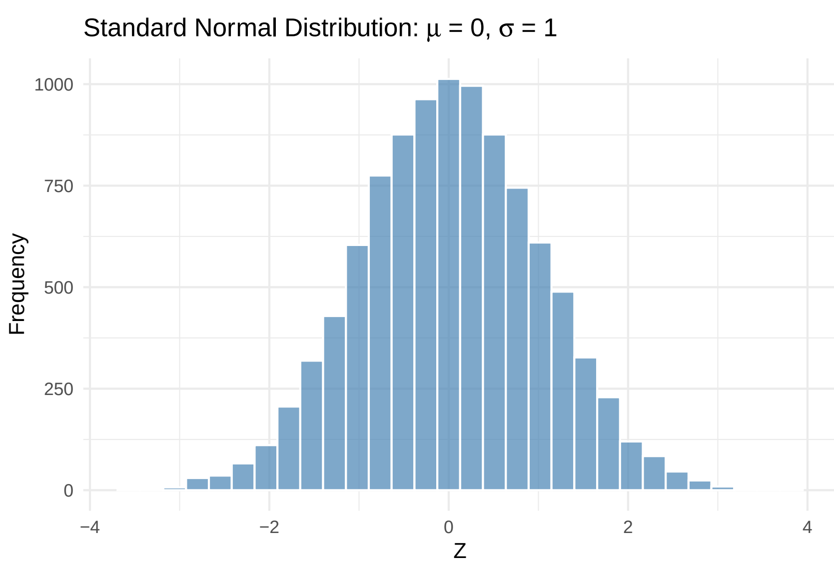
$$Z \sim N(0, 1)$$

The standard normal is important because any normal random variable can be converted to a standard normal through **standardization**. If $X \sim N(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Standardization subtracts the mean (centering the distribution at zero) and divides by the standard deviation (scaling it to have unit variance).

Figure B.3: A histogram of 10,000 draws from a standard normal distribution (mean 0, standard deviation 1).



The standard normal is what z-tables (or statistical software) use to compute probabilities. Once you standardize a variable, you can look up the probability of observing values in any range.

B.6 Summary

This appendix reviewed the mathematical and statistical foundations needed for econometrics. We covered the summation operator and its properties, which are essential for understanding formulas throughout the course. We introduced derivatives and partial derivatives, which help us interpret how changes in one variable affect another, and learned how to find and classify maxima and minima of functions. We distinguished between populations and samples—a fundamental concept for statistical inference. We then turned to probability, defining random variables, expected values, variance, covariance, and correlation. Finally, we introduced the normal distribution, which plays a central role in statistical inference.

B.6.1 Key Formulas

Concept	Formula
Sample mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Critical point (quadratic)	$x^* = -\frac{b}{2a}$ for $f(x) = ax^2 + bx + c$
Second derivative test	Max if $f''(x^*) < 0$; Min if $f''(x^*) > 0$
Expected value	$E[X] = \sum_{j=1}^{k} x_j f(x_j) = \mu$
Variance	$\text{Var}(X) = E[(X-\mu)^2] = \sigma^2$
Standard deviation	$\text{sd}(X) = \sqrt{\text{Var}(X)} = \sigma$
Covariance	$\text{Cov}(X, Y) = E[(X-\mu_X)(Y-\mu_Y)]$
Correlation	$\rho_{XY} = \frac{\text{Cov}(X,Y)}{\text{sd}(X)\cdot\text{sd}(Y)}$
Standardization	$Z = \frac{X-\mu}{\sigma}$

B.7 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.